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Fawcett et al.

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(54) **ELECTRONIC LOCKS FOR SERVER RACKS**

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(51) **Int. Cl.**

H05K 5/02	(2006.01)
G07C 9/00	(2020.01)
H05K 7/14	(2006.01)

(52) **U.S. Cl.**
CPC ***H05K 5/0221*** (2013.01); ***G07C 9/00309*** (2013.01); ***G07C 9/00857*** (2013.01);
(Continued)

(58) **Field of Classification Search**
None
See application file for complete search history.

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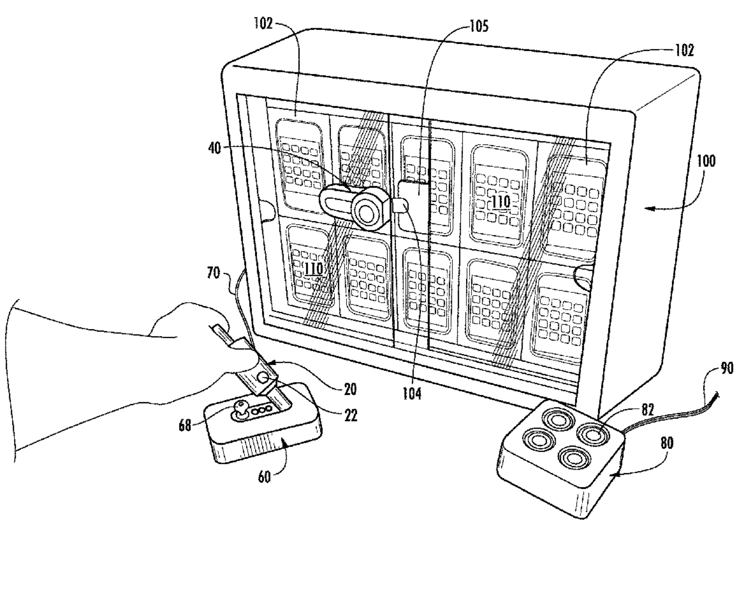
(57) **ABSTRACT**

Embodiments of the present invention are directed to systems and methods for preventing unauthorized access to server racks. In one example, a system includes an electronic key and an electronic lock configured to be attached to the server rack. The electronic lock is configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the lock. The electronic lock includes a handle configured to be disengaged from the electronic lock when the electronic key is authorized, and the handle is configured to be actuated to unlock the lock for accessing the server rack when the handle is disengaged from the electronic lock. The electronic lock is configured to re-lock only when the handle is engaged with the electronic lock.

19 Claims, 28 Drawing Sheets



(52) **U.S. Cl.**
CPC . ***H05K 7/1495*** (2013.01); *G07C 2009/00388* (2013.01); *G07C 2009/00579* (2013.01); *G07C 2009/00634* (2013.01)

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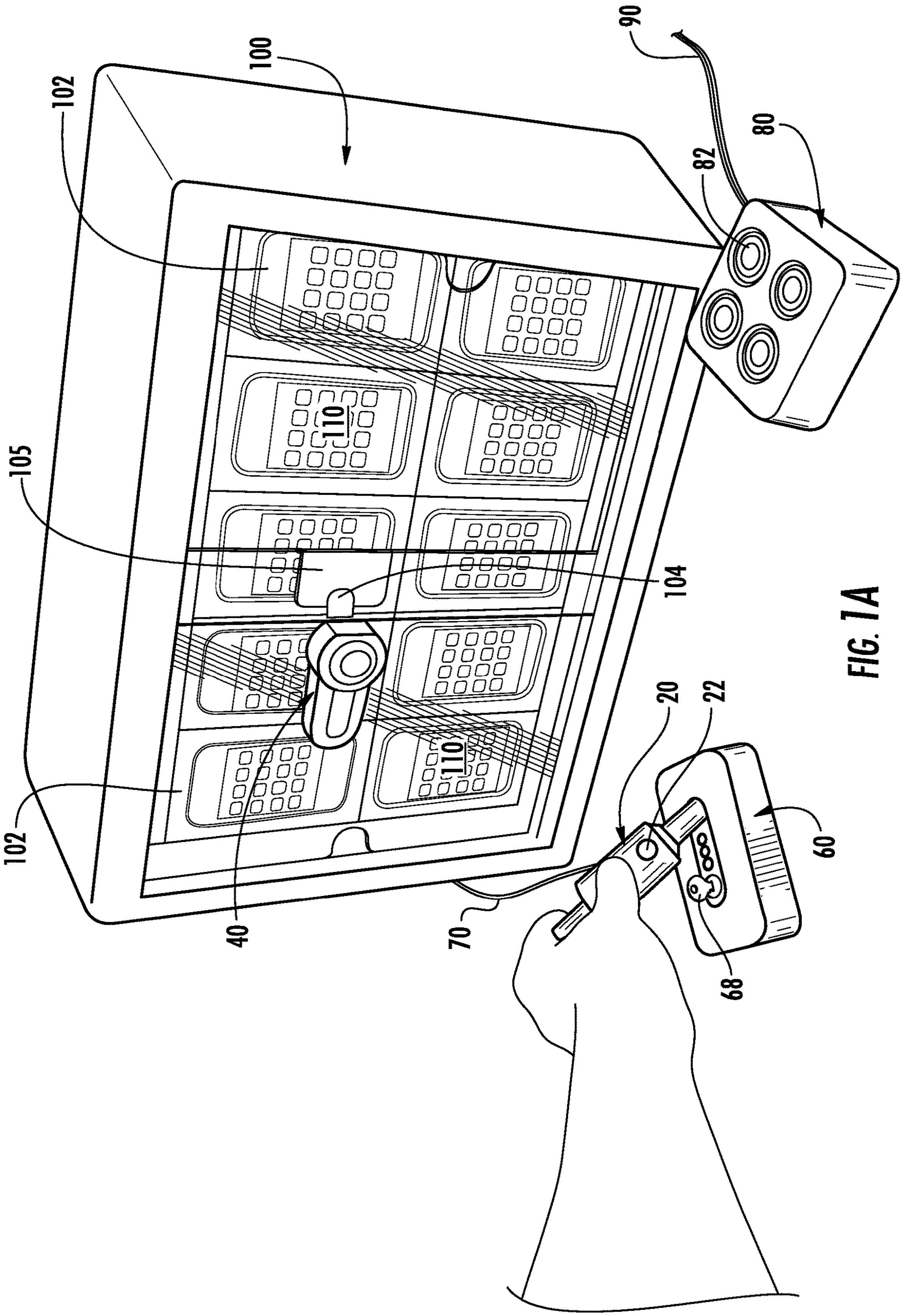

FIG. 1A

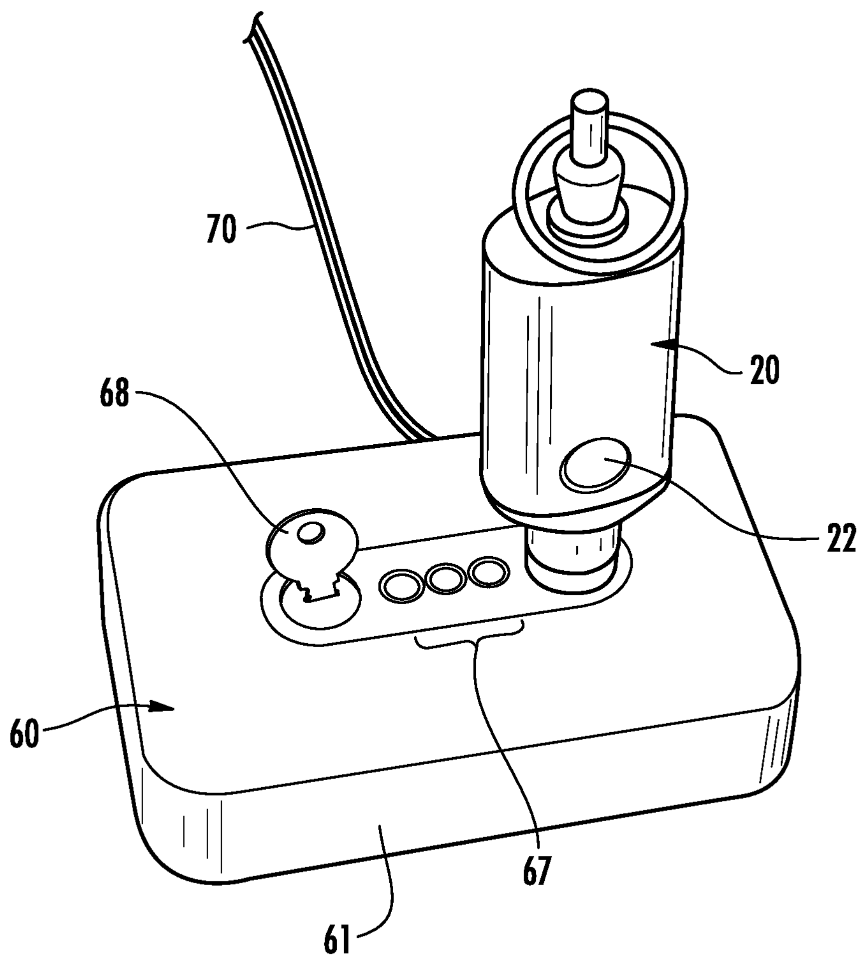

FIG. 1B

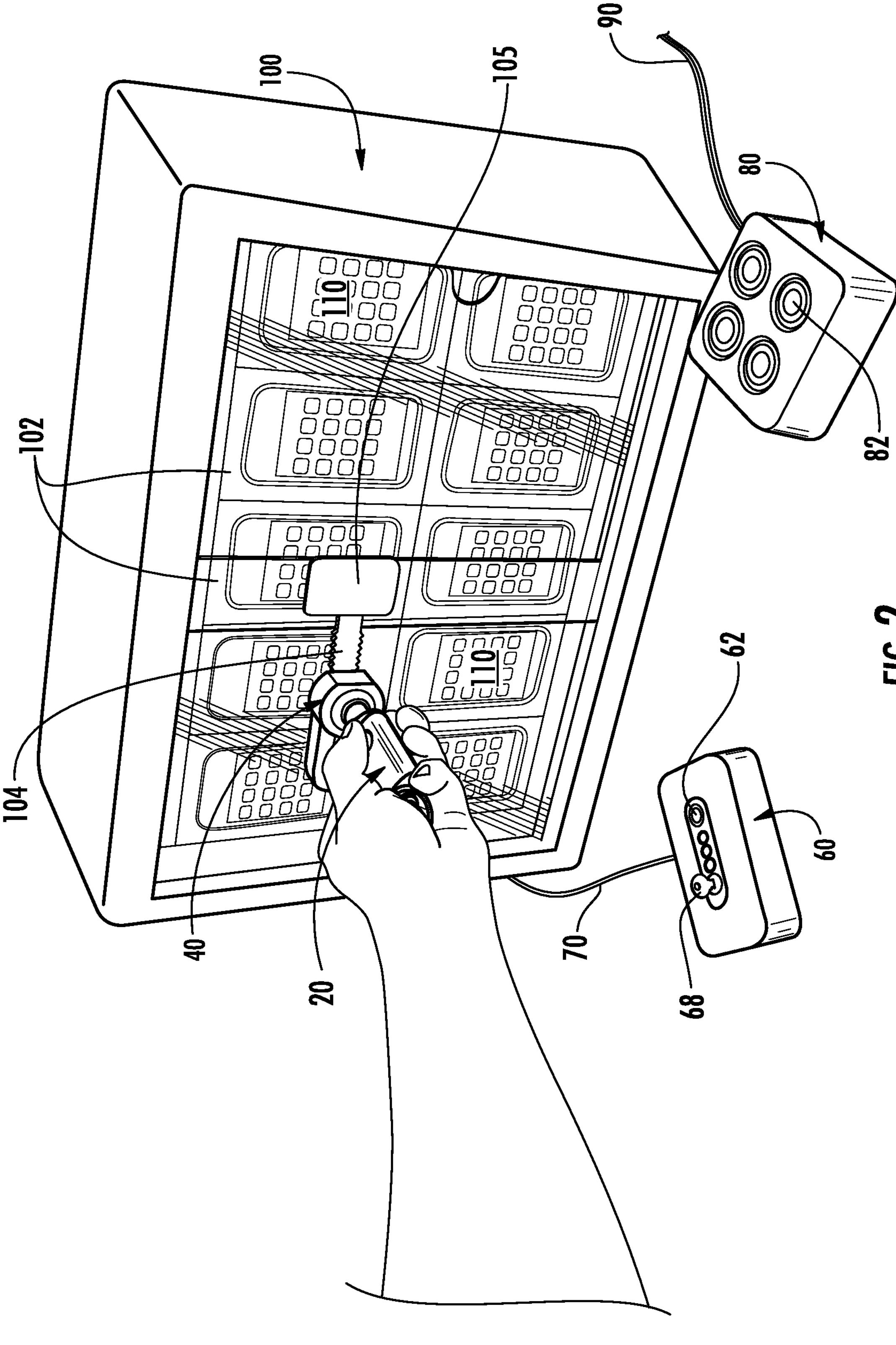

FIG. 2

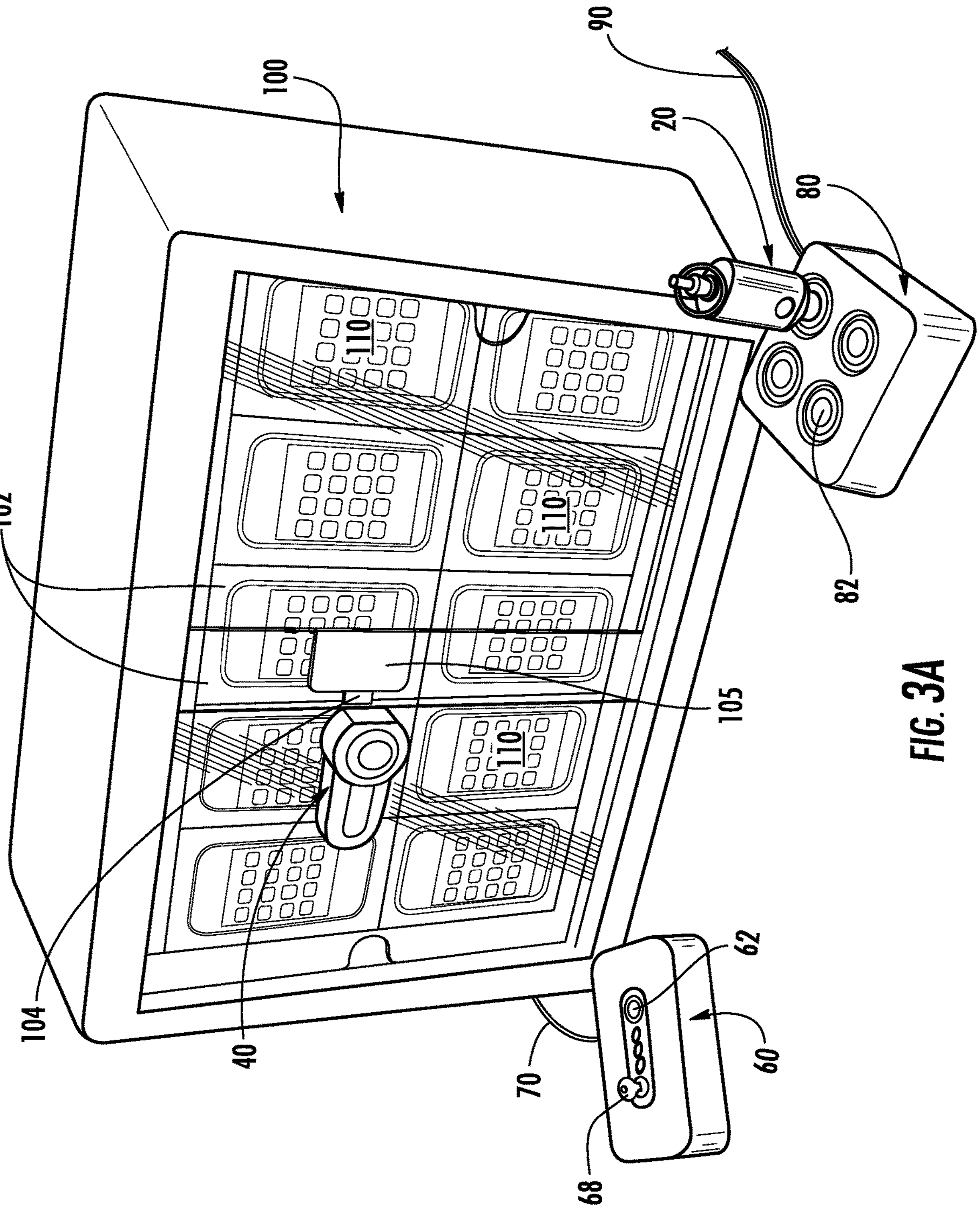

FIG. 3A

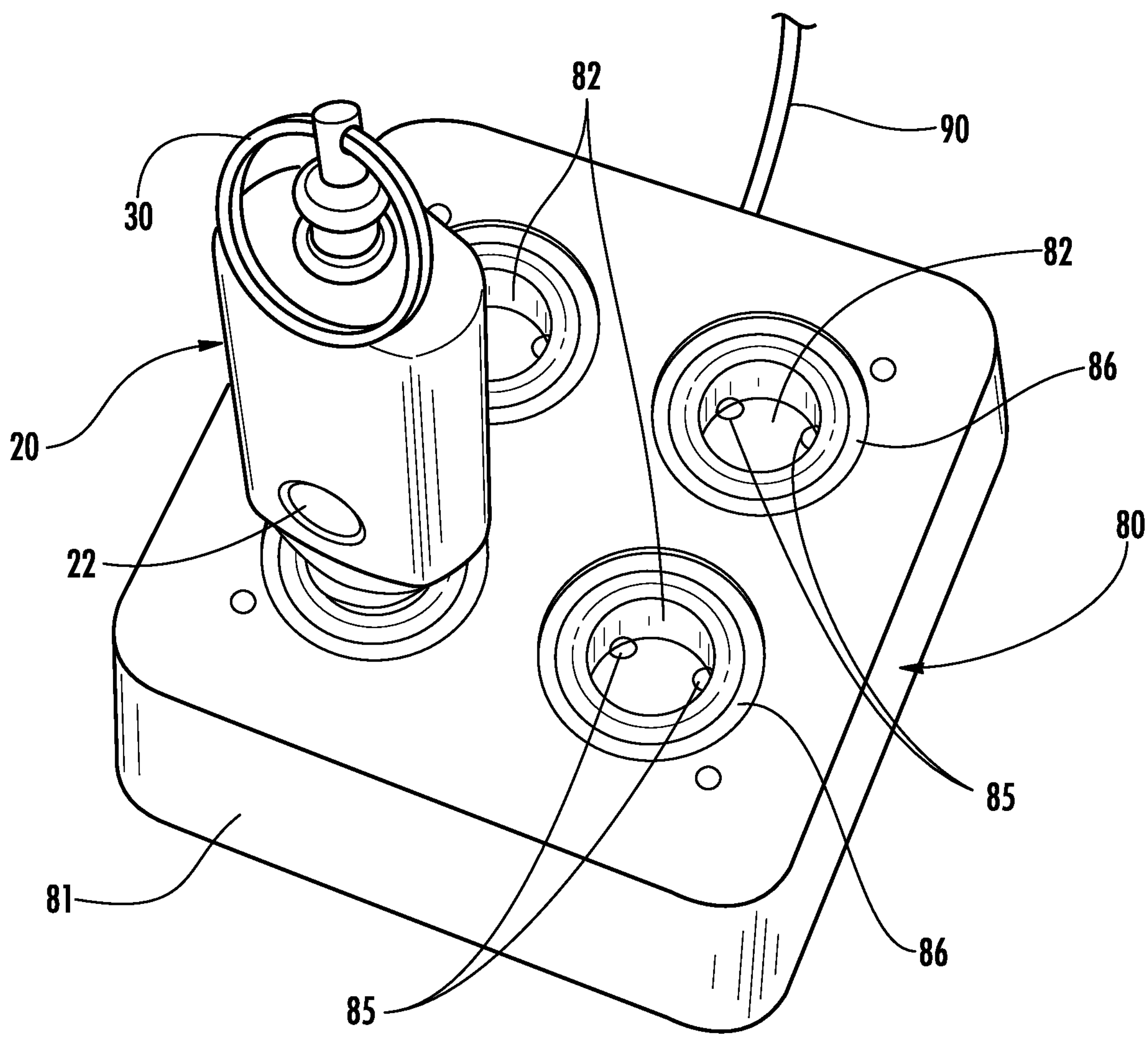

FIG. 3B

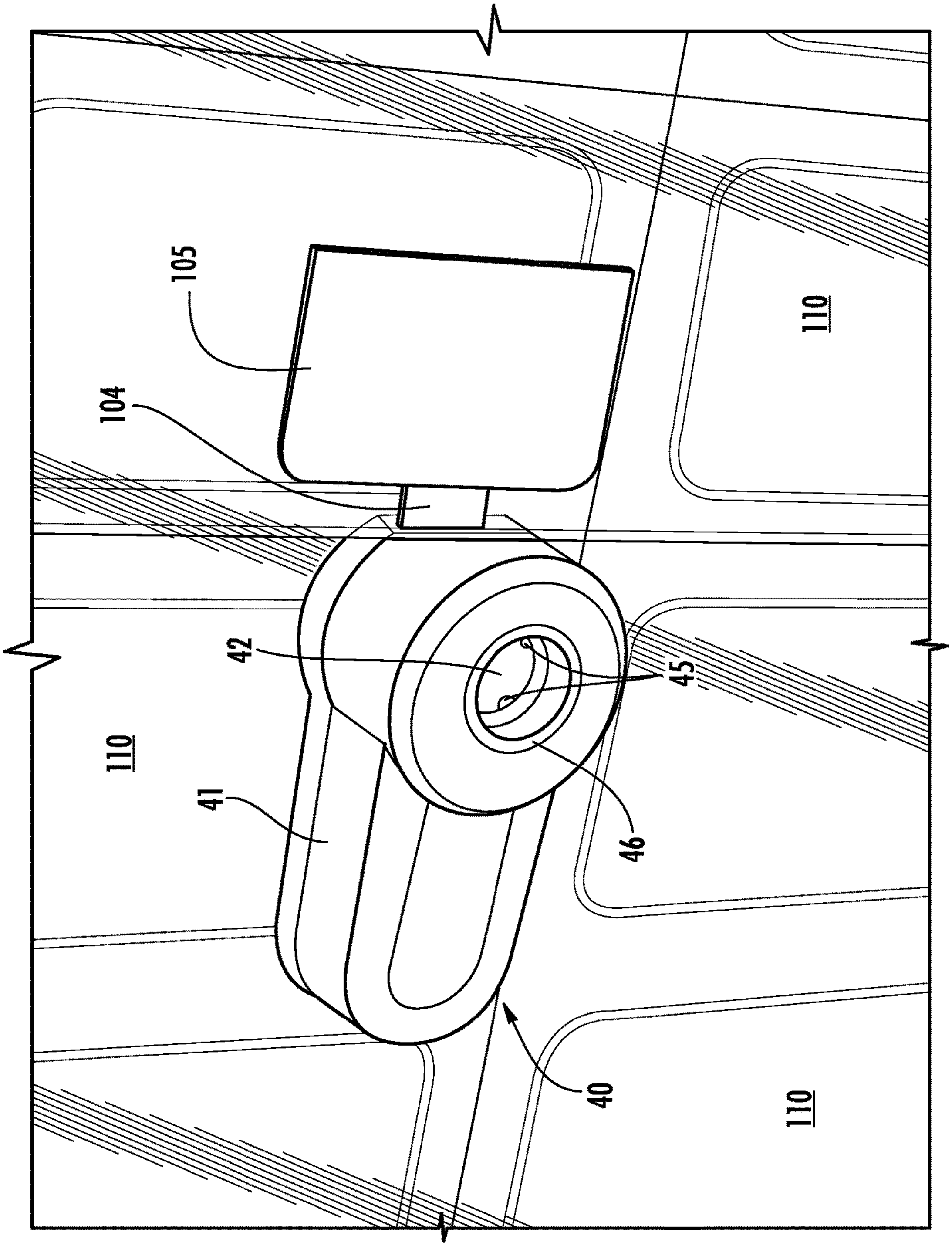

FIG. 4

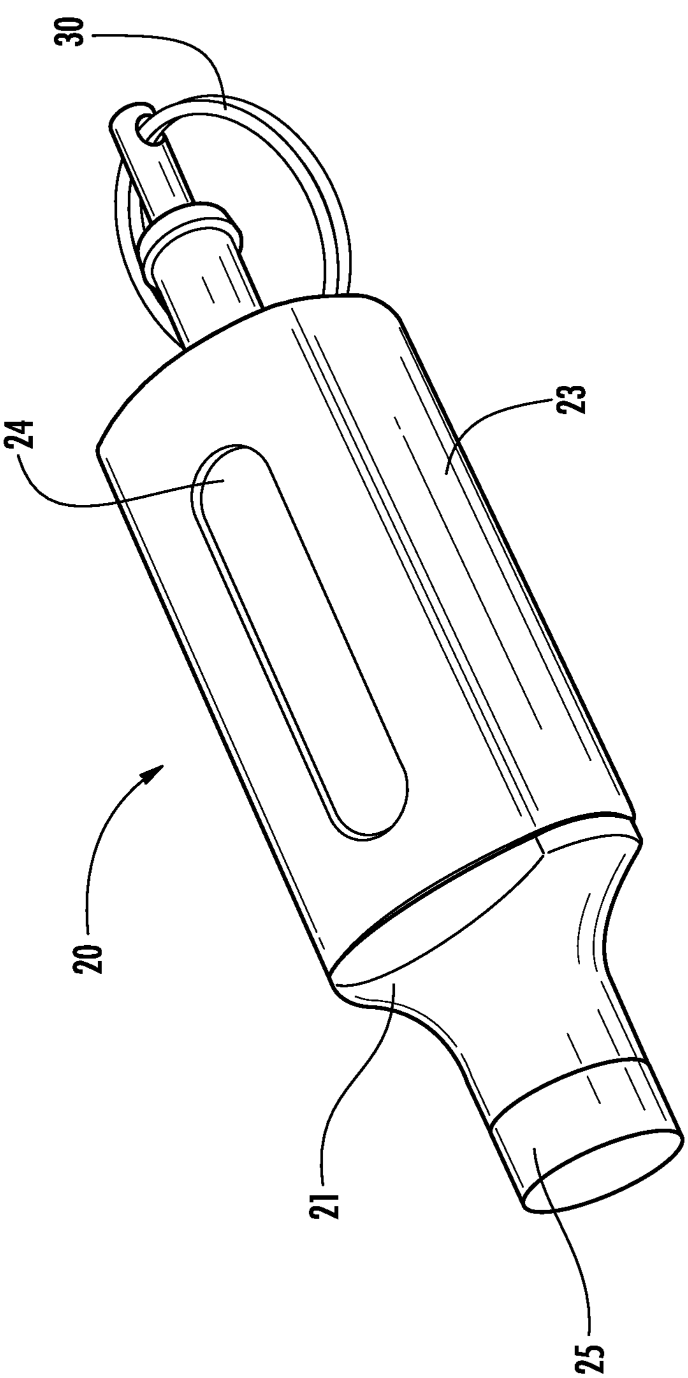

FIG. 5

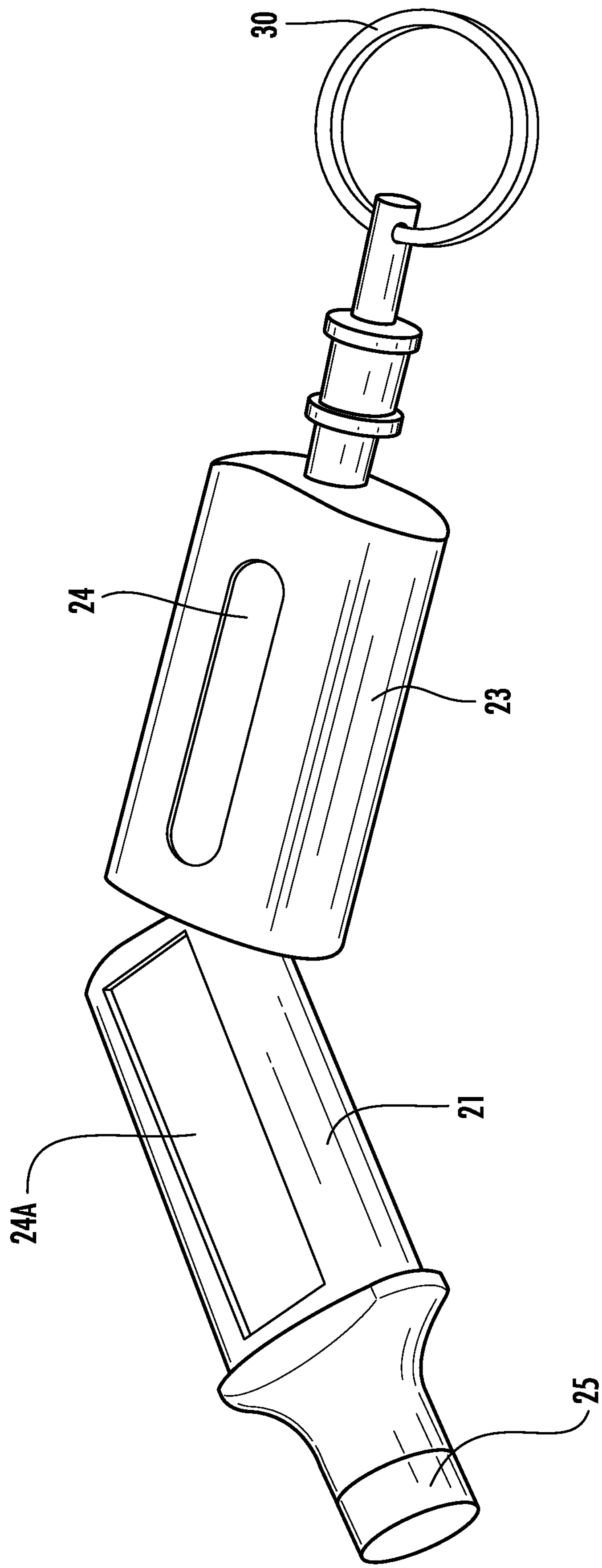

FIG. 6

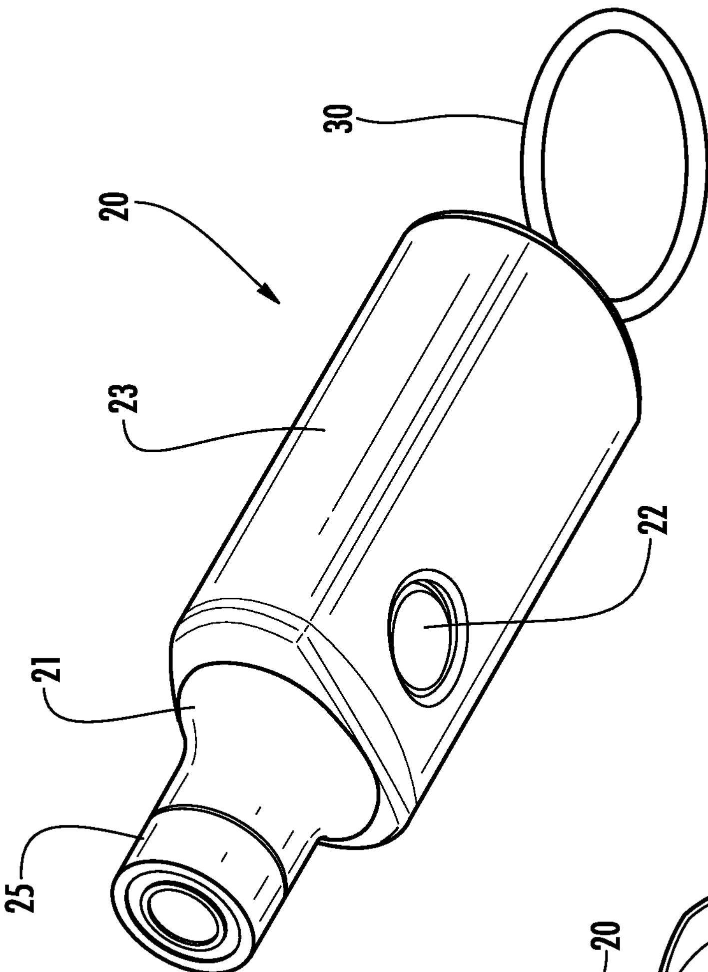

FIG. 7A

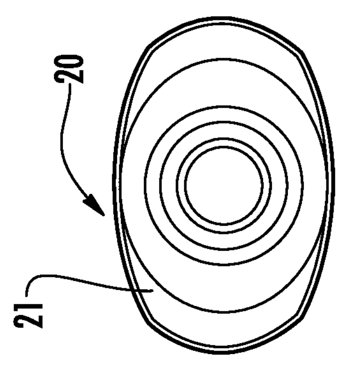

FIG. 7B

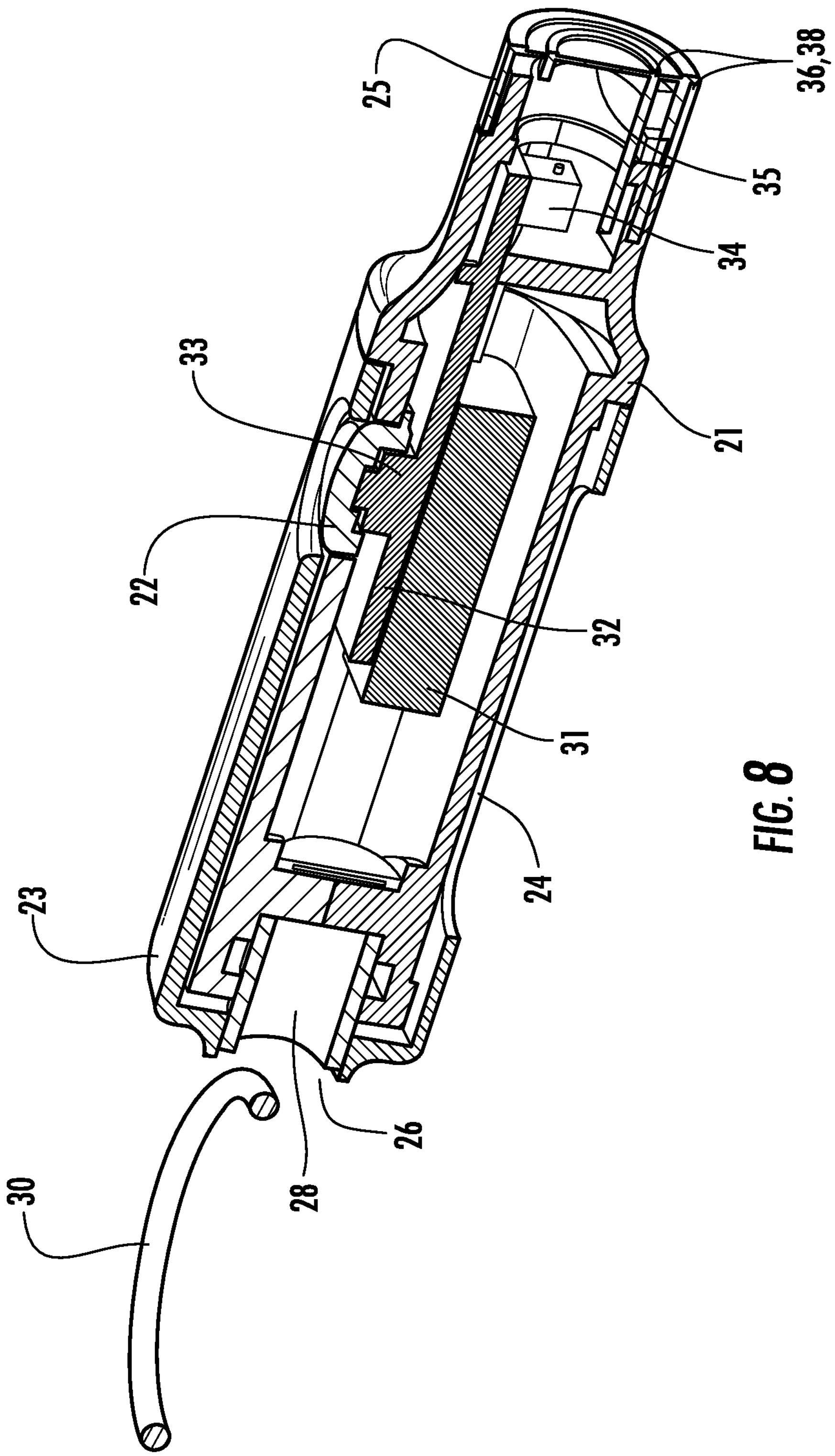

FIG. 8

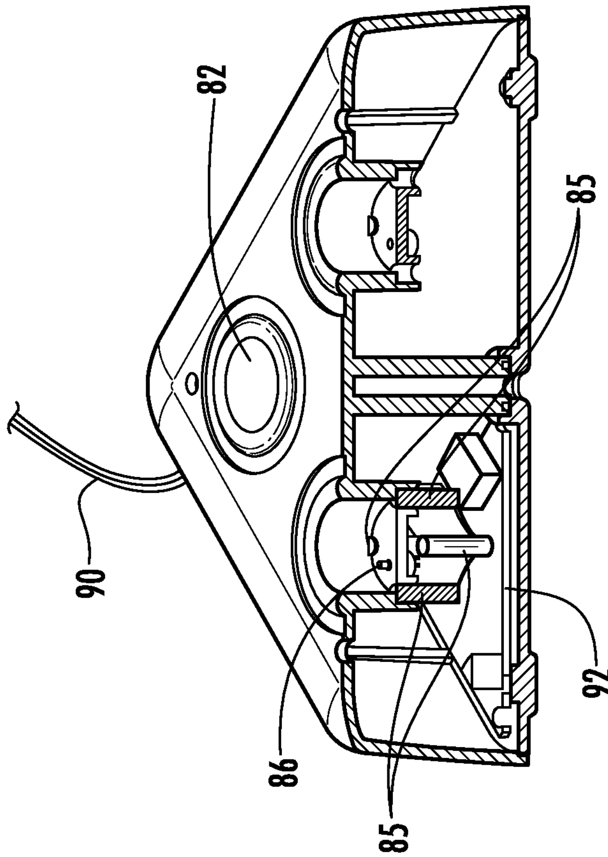

FIG. 9B

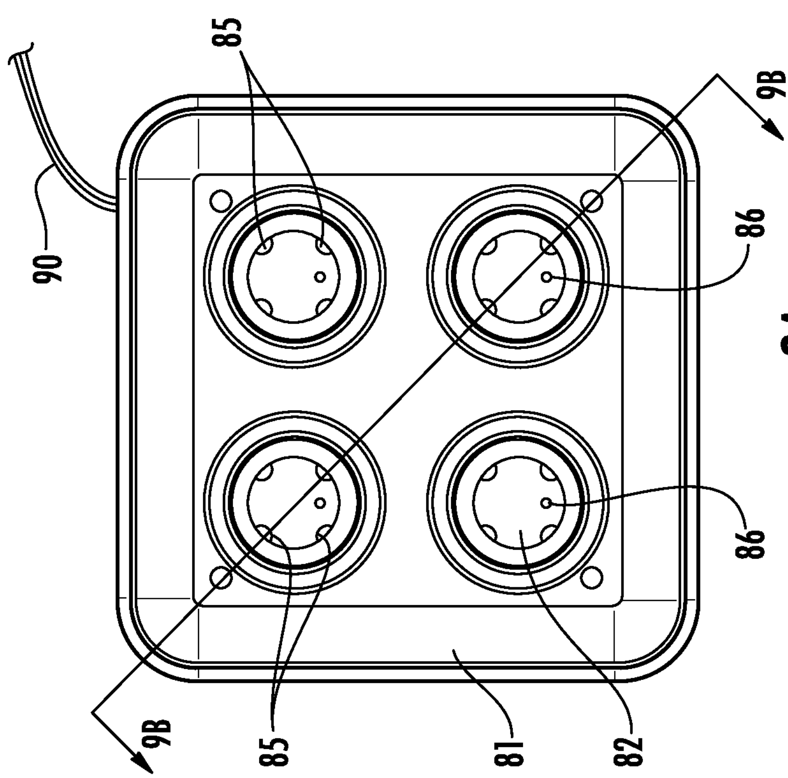

FIG. 9A

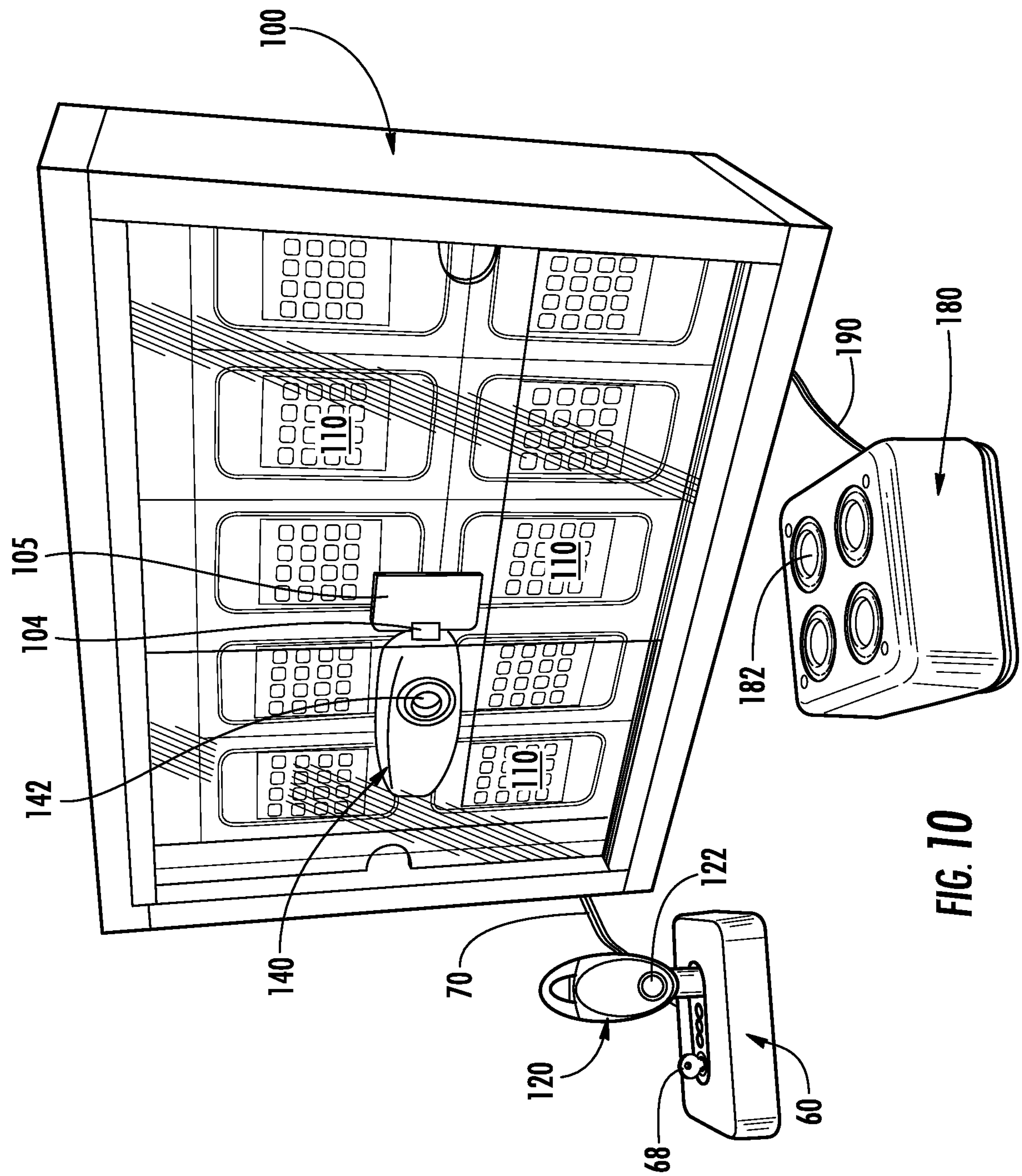

FIG. 10

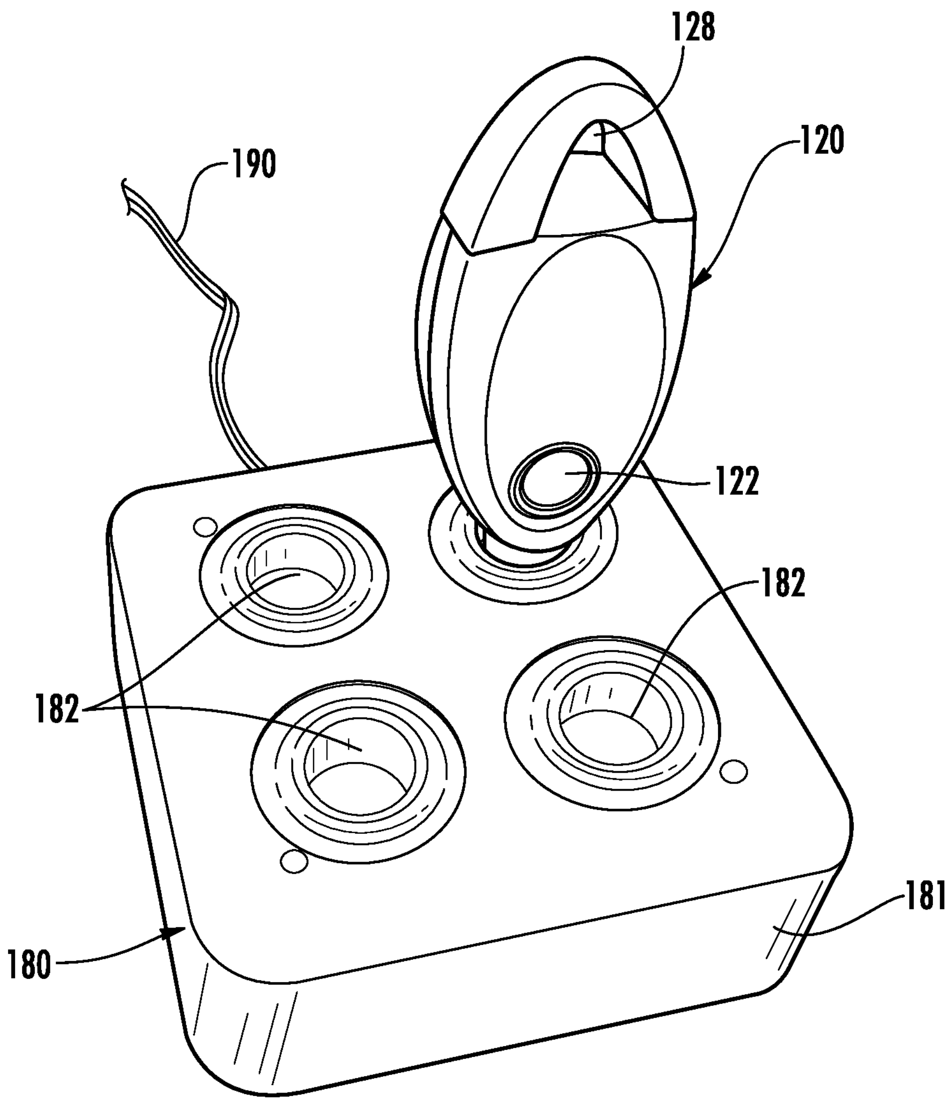

FIG. 11

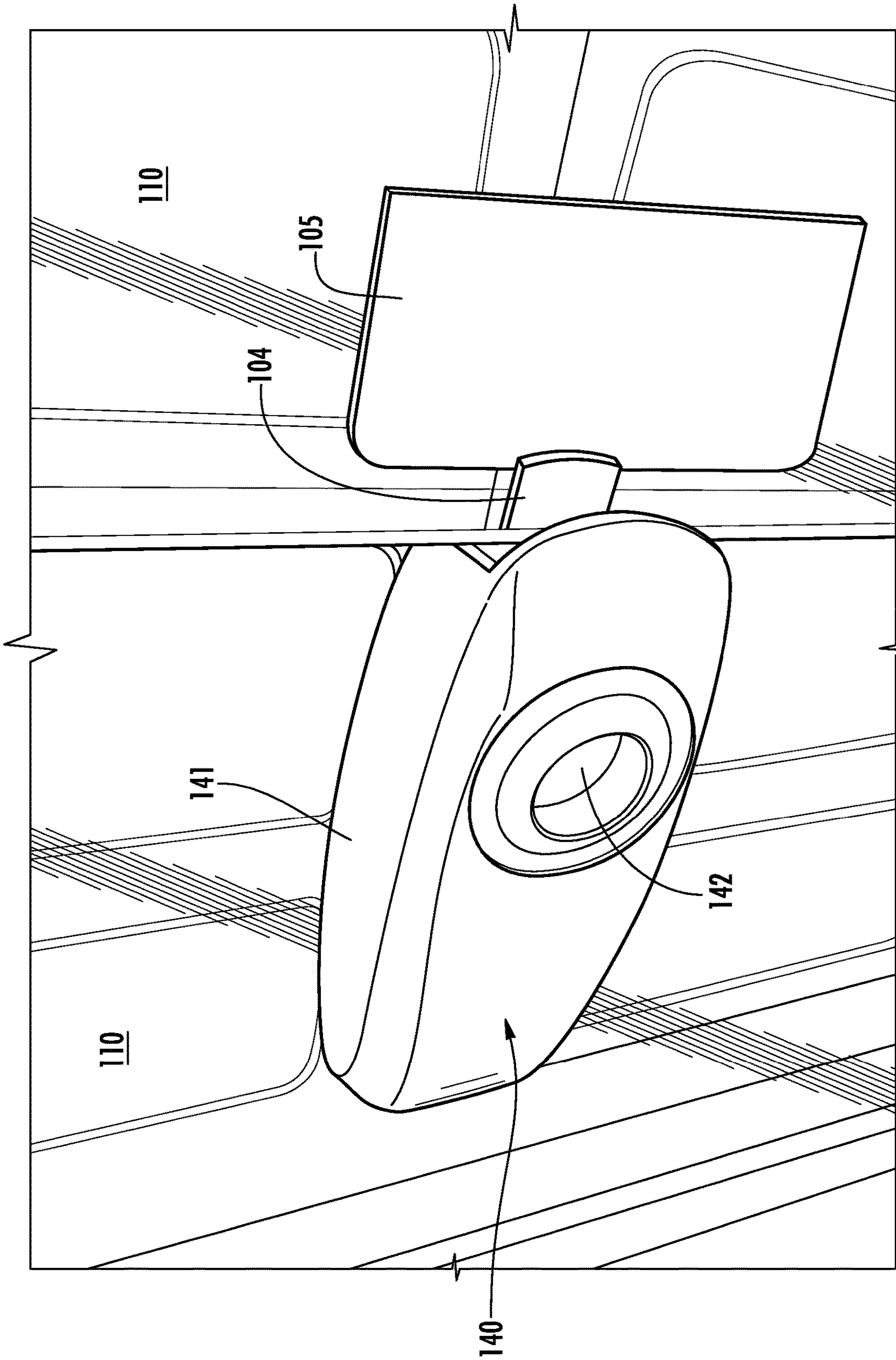

FIG. 12

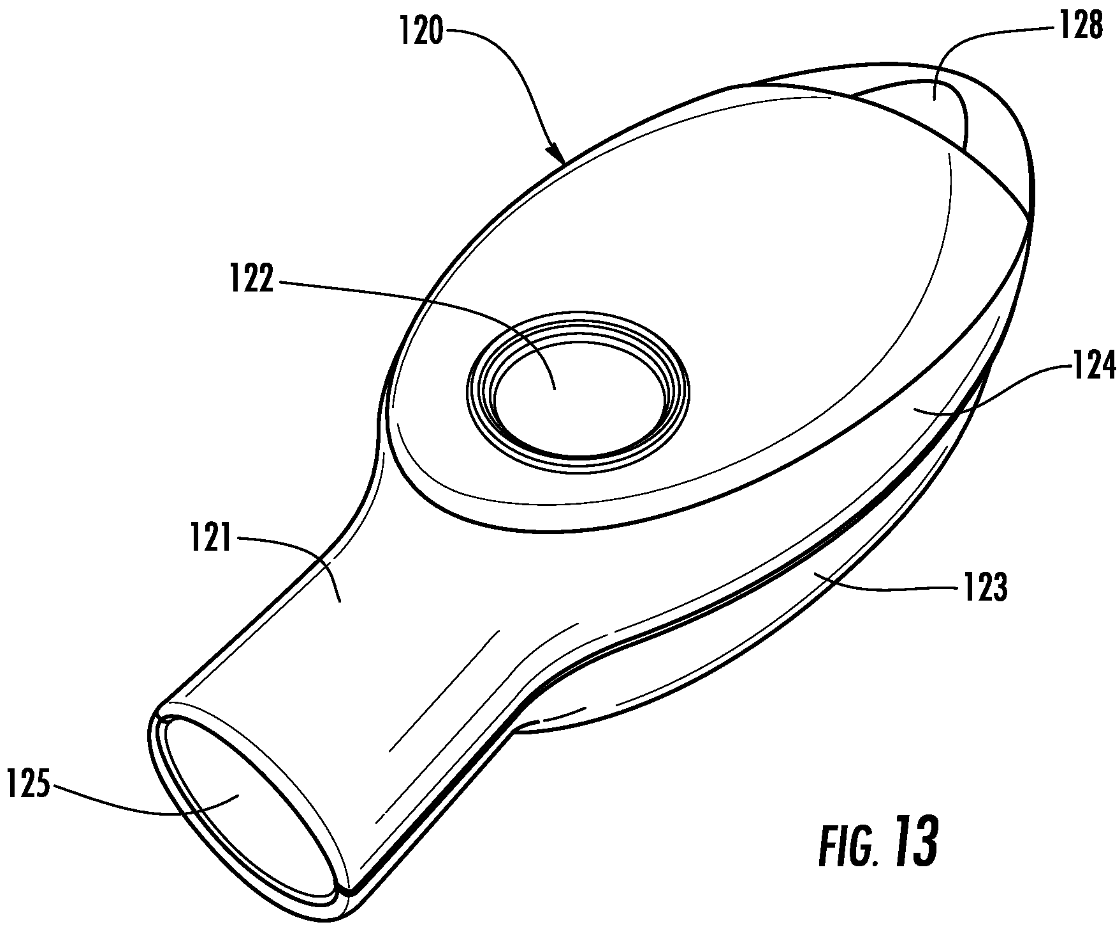

FIG. 13

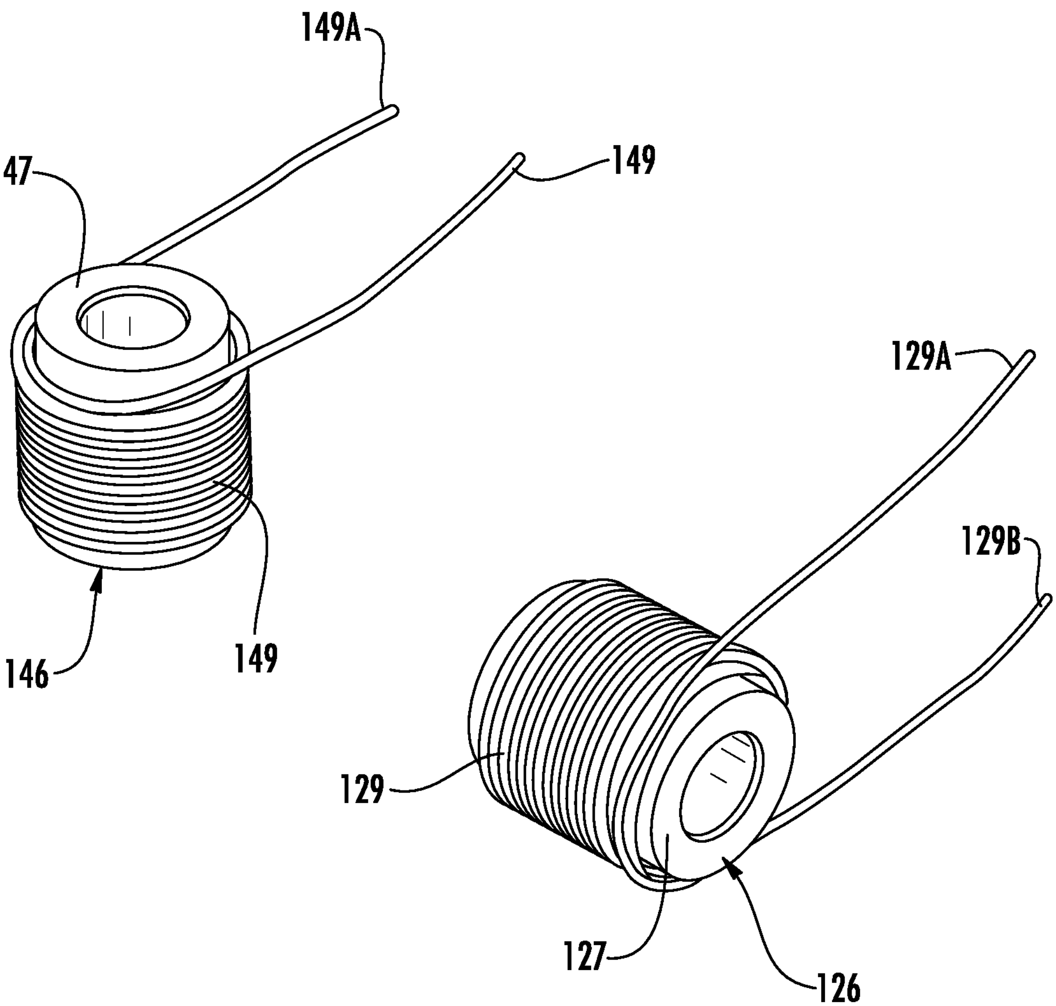

FIG. 14

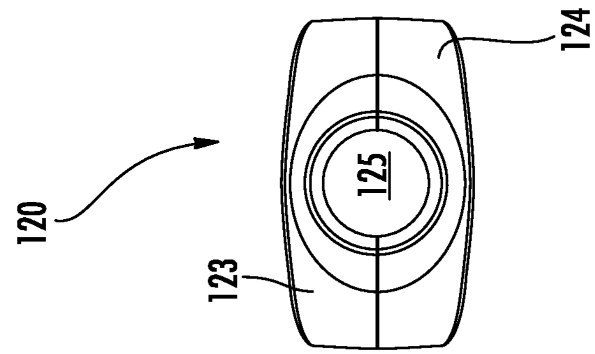

FIG. 15B

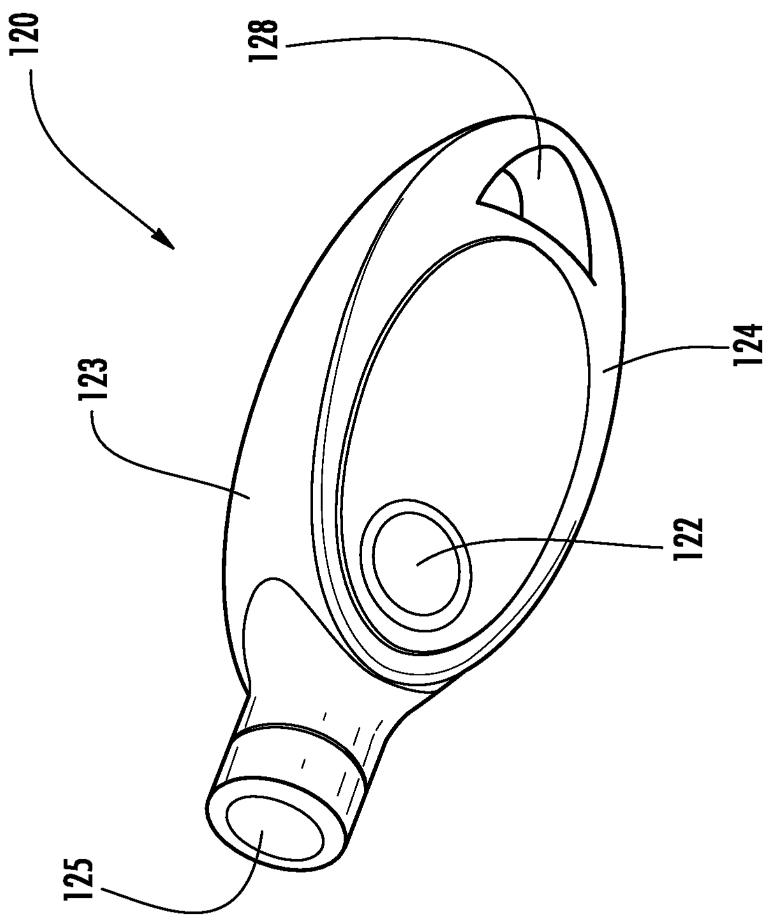

FIG. 15A

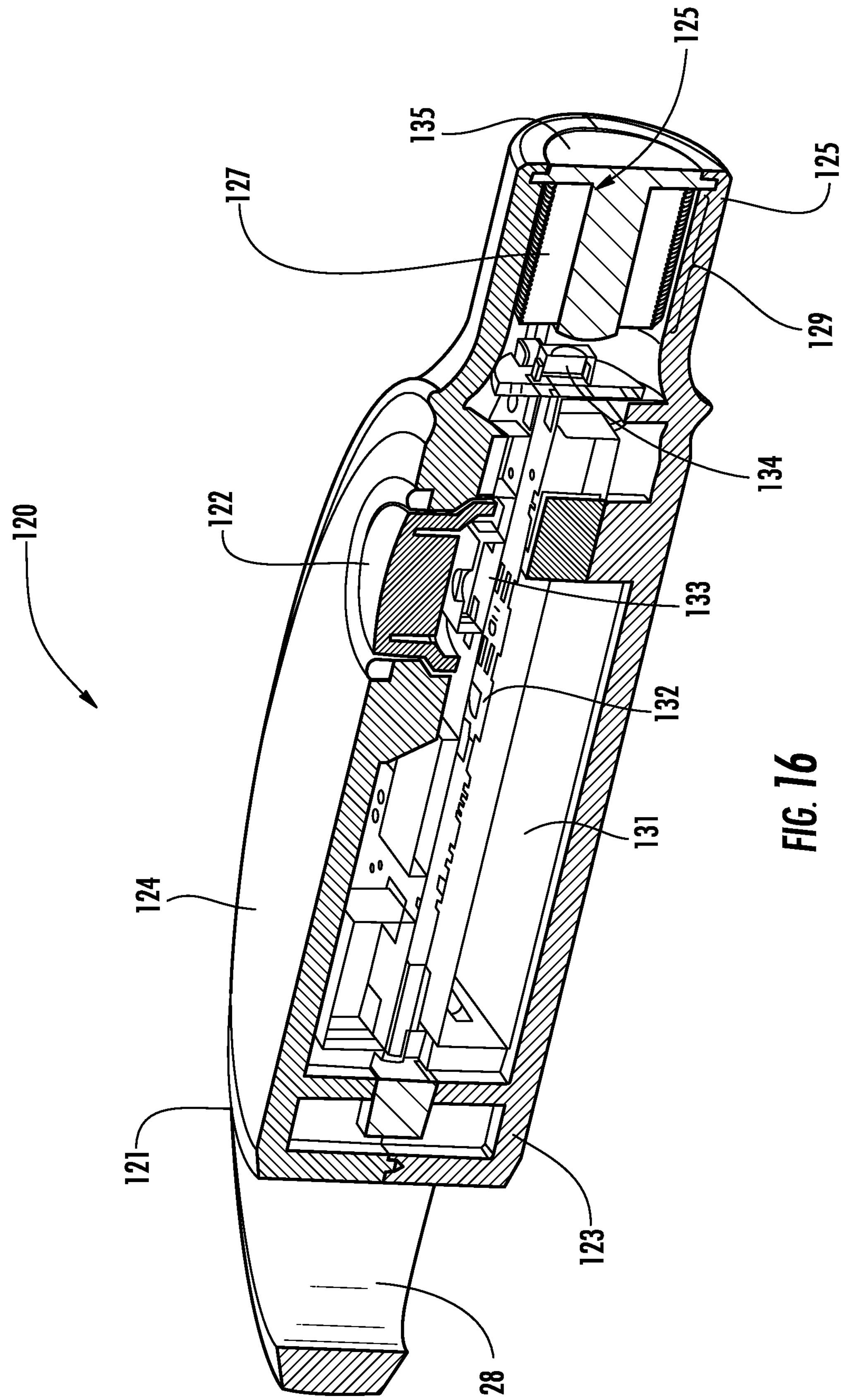

FIG. 16

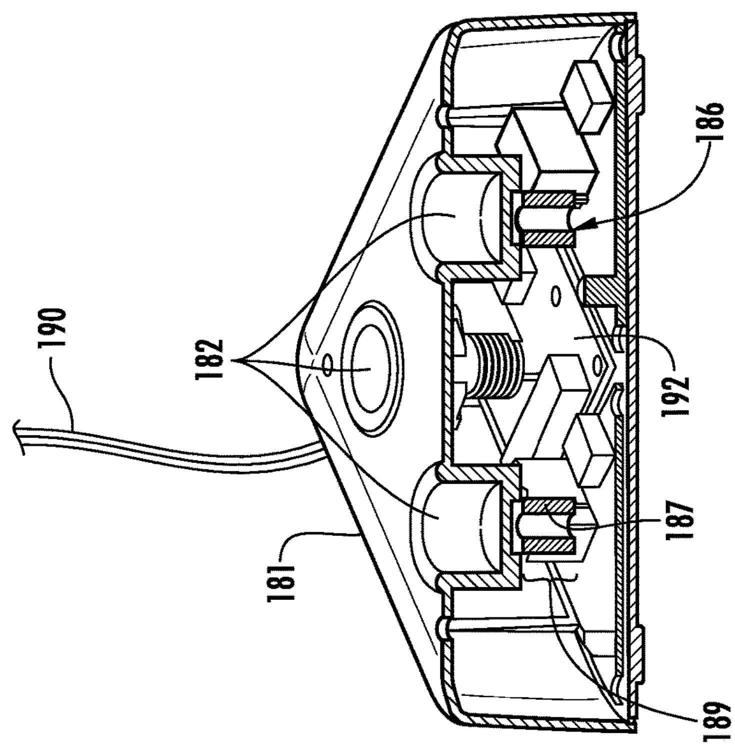

FIG. 17B

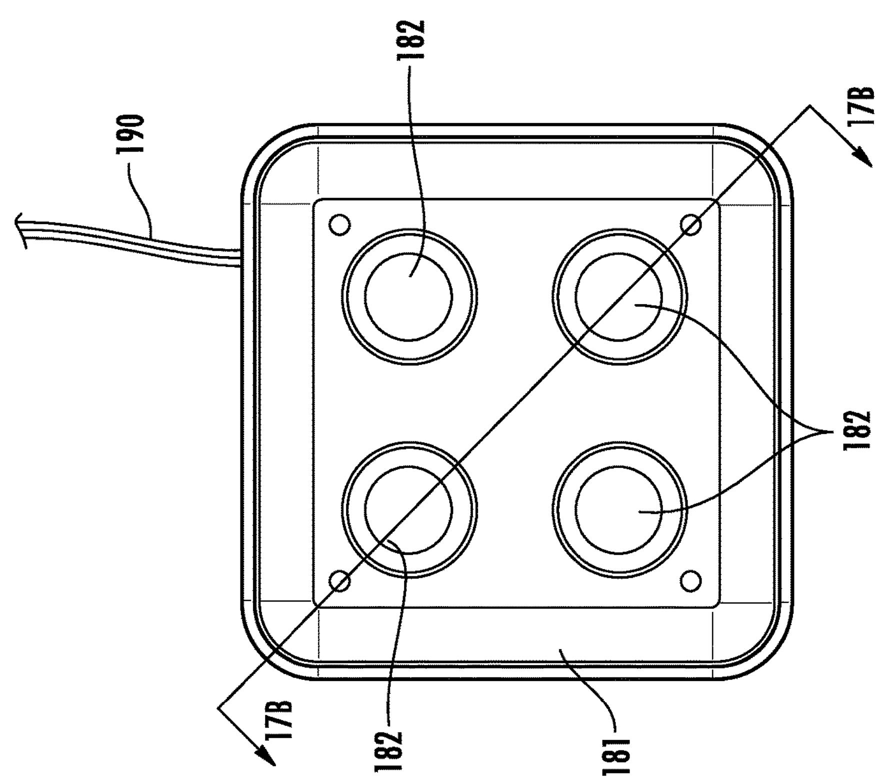

FIG. 17A

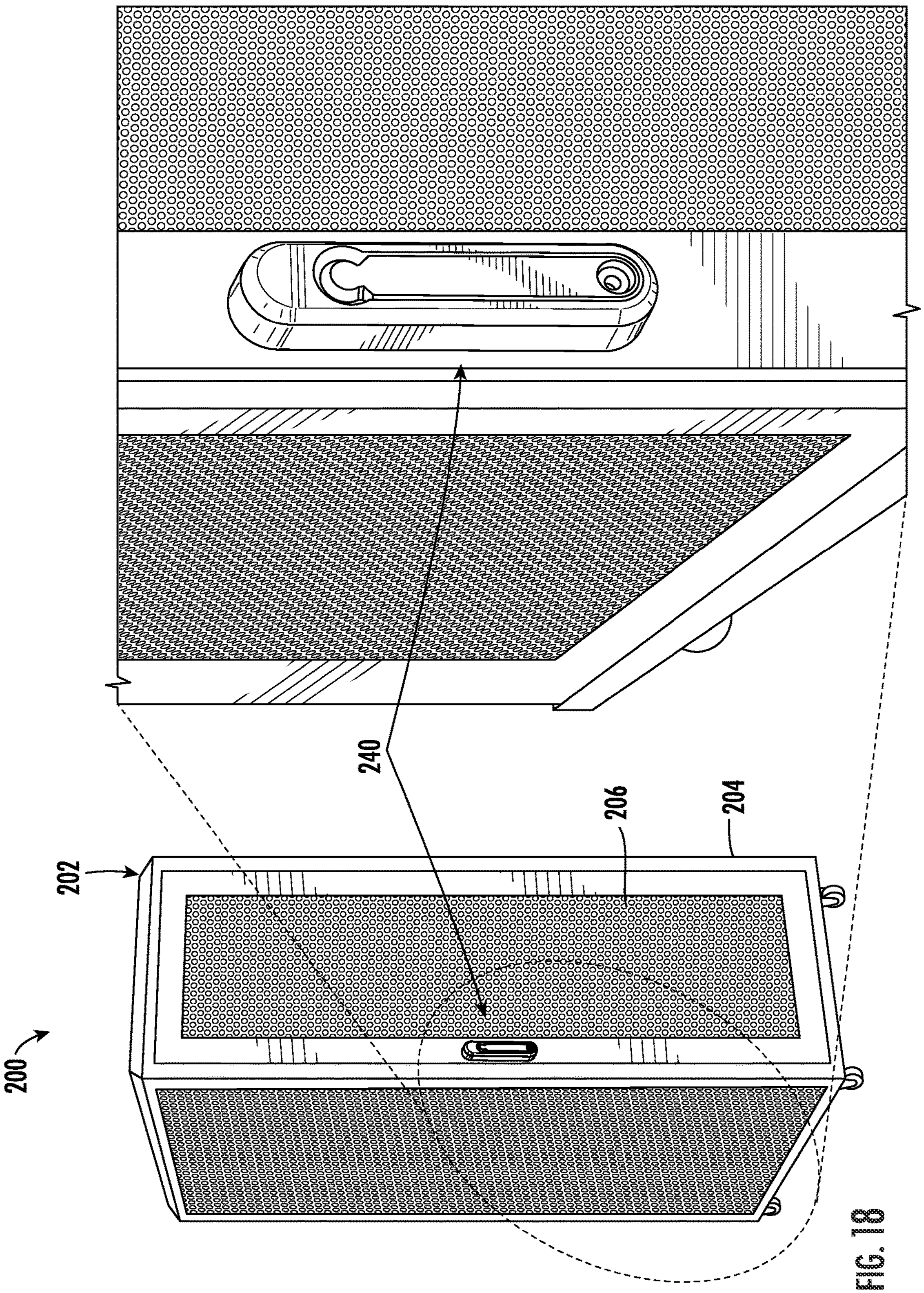

FIG. 18

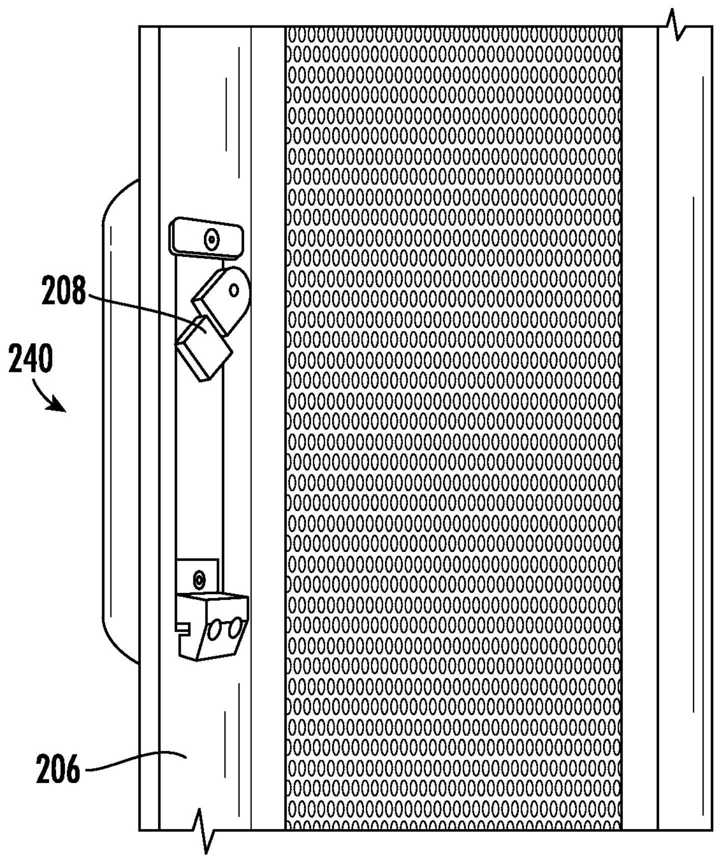

FIG. 19

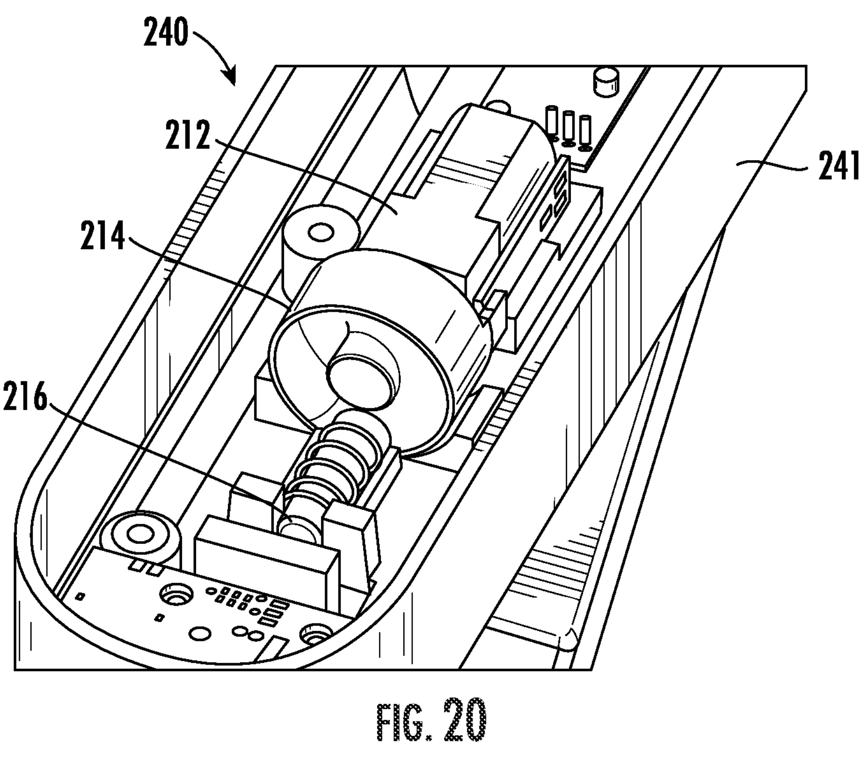

FIG. 20

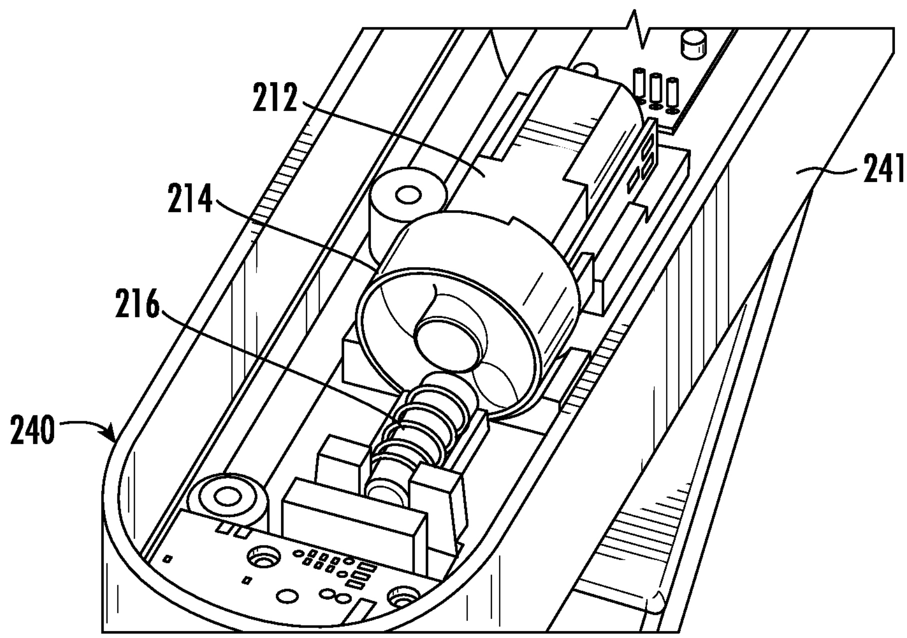

FIG. 21

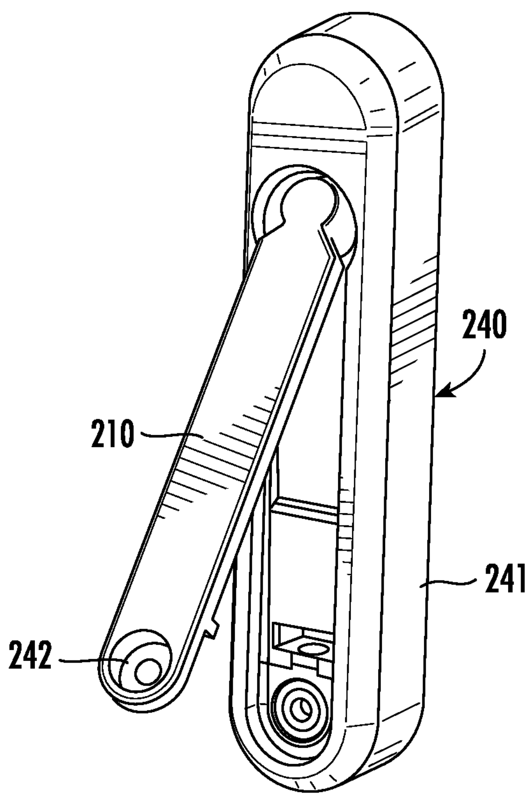

FIG. 22

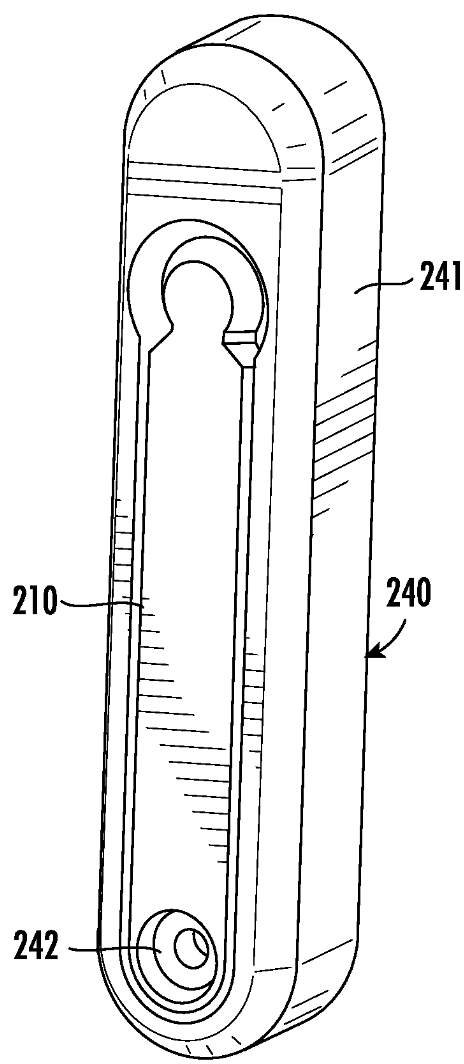

FIG. 23

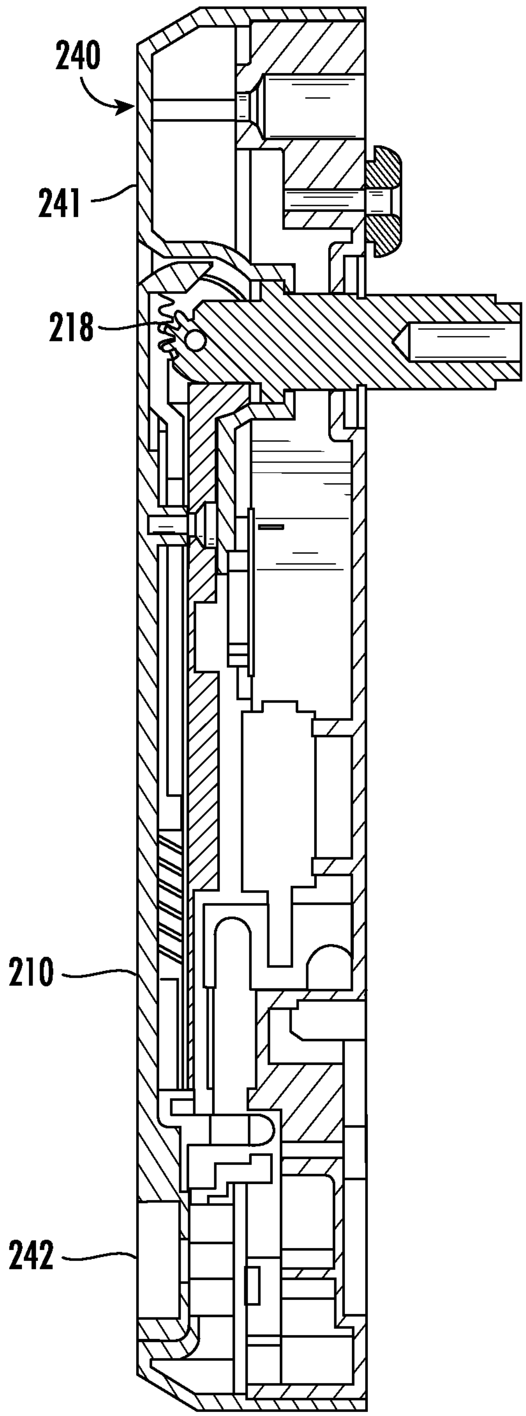

FIG. 24

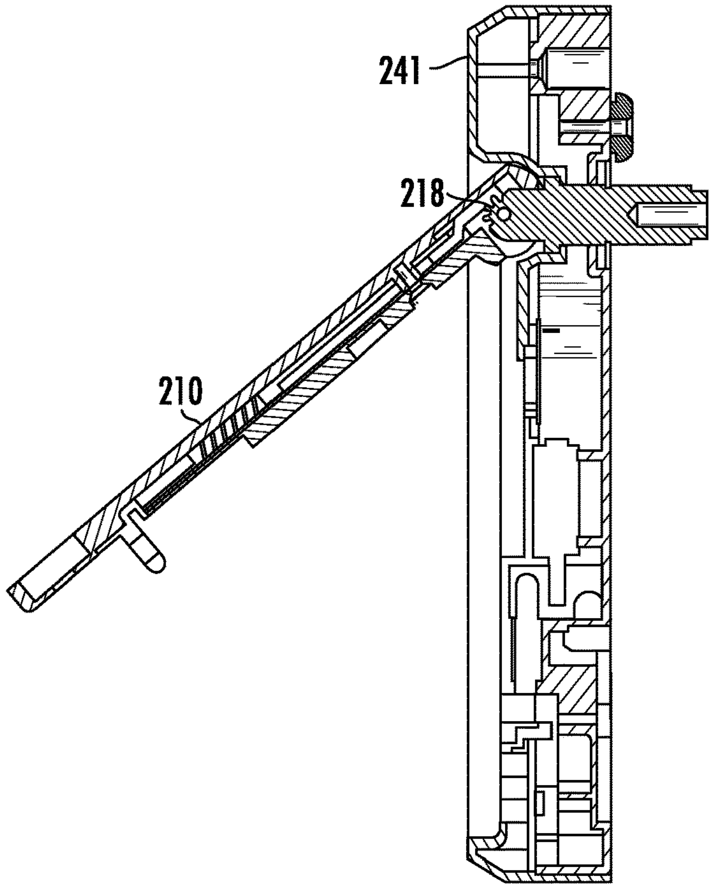

FIG. 25

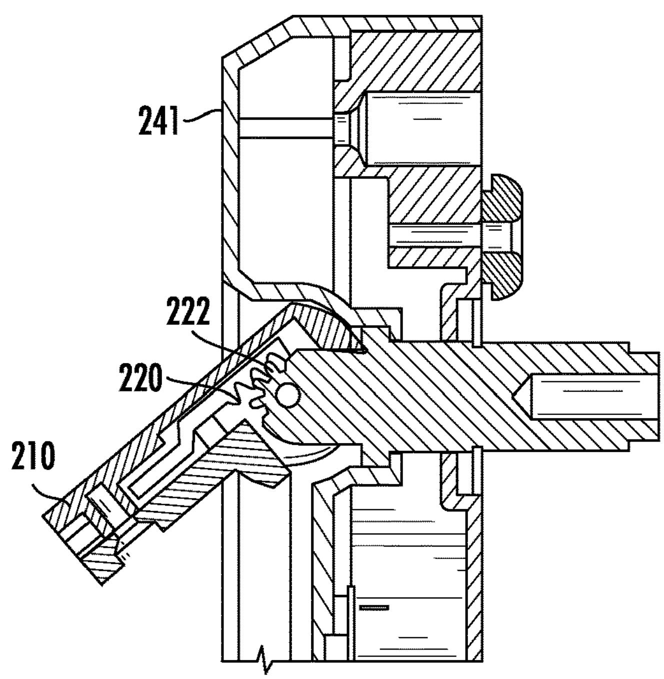

FIG. 26

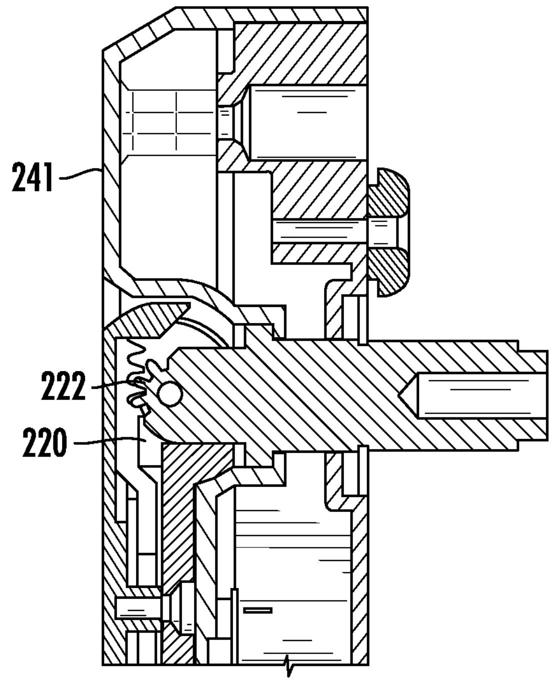

FIG. 27

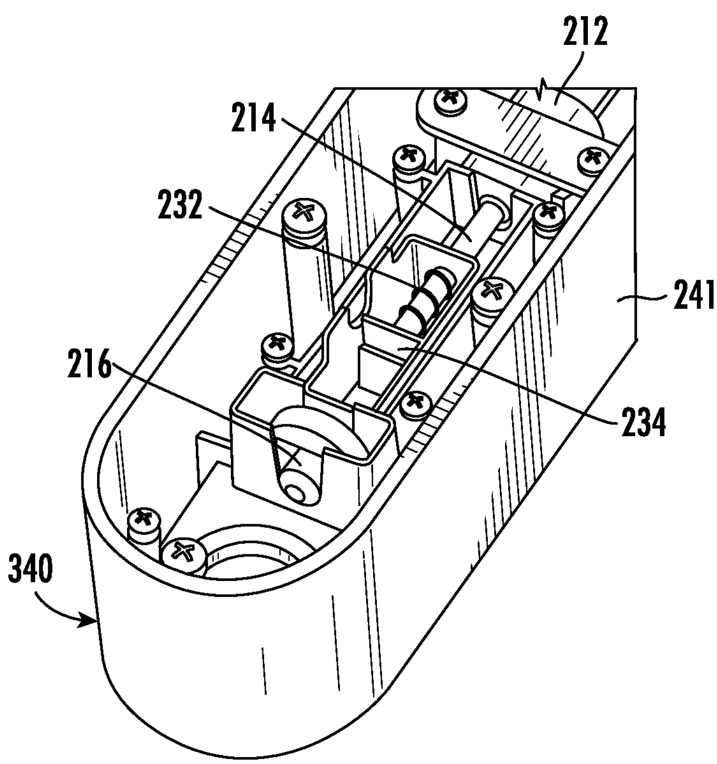

FIG. 28

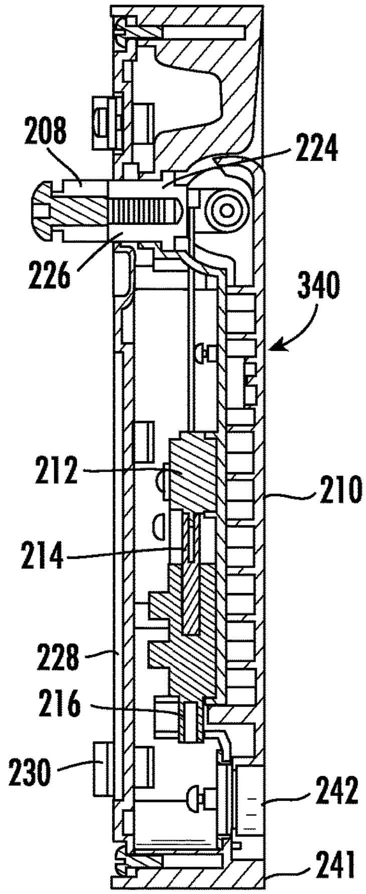
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FIG. 29

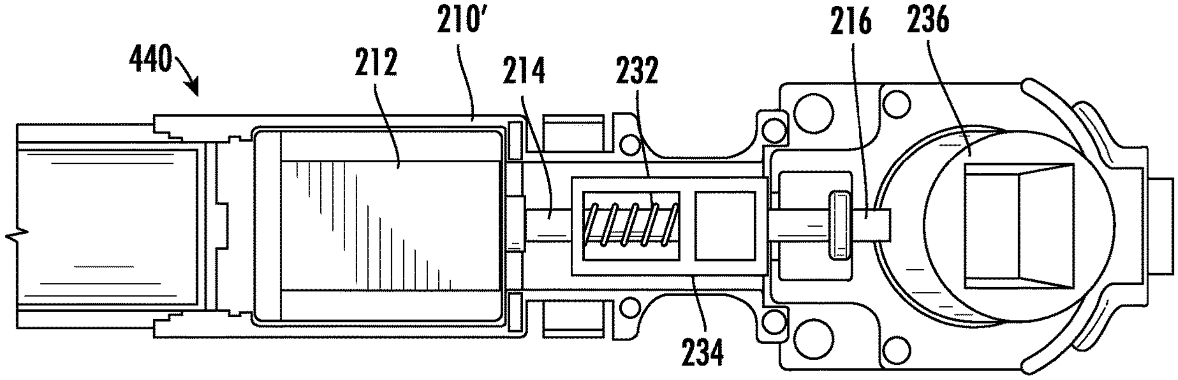
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FIG. 30

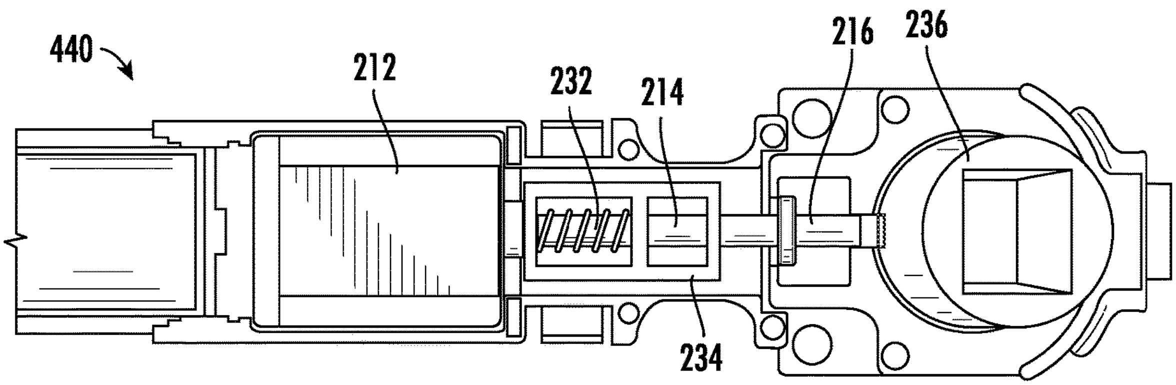

FIG. 31

ELECTRONIC LOCKS FOR SERVER RACKS

CROSS REFERENCE TO RELATED APPLICATIONS

This application claims the benefit of priority to and is a 371 U.S. national phase entry of International Application No. PCT/US2021/70993, filed on Jul. 27, 2021, which is a non-provisional of and claims the benefit of priority to U.S. Provisional Application No. 63/057,580, filed on Jul. 28, 2020 and U.S. Provisional. No. 63/059,280, filed on Jul. 31, 2020, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

Embodiments of the present invention relates generally to electronic locks, systems, and methods for server racks.

BACKGROUND OF THE INVENTION

Server racks are generally protected in the market by standard mechanical keys and/or combination codes which have issues such as broken keys, ease of copying, difficulty in managing access to multiple locks with multiple keys and multiple users, and no traceability to show who accessed the racks and when. Electronic locks address some of the issues with mechanical keys but typically require cumbersome wiring and infrastructure overhead in order to provide power and control for each lock. In addition, in order to electronically track any user interactions with electronic locks, an extensive network of data cables and hubs also need to be installed.

BRIEF SUMMARY

Embodiments of the present invention are directed towards a security system for a server rack comprising an electronic key and an electronic lock configured to be attached to the server rack. The electronic lock is configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock. The electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized, wherein the handle is configured to be actuated to unlock the lock for accessing the server rack when the handle is disengaged from the electronic lock, and wherein the electronic lock is configured to re-lock only when the handle is engaged with the electronic lock.

In another embodiment, a security system for a server rack comprises a server rack comprising a cabinet and a door. The security system includes an electronic key and an electronic lock configured to be attached to the server rack. The electronic lock is configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock. In some aspects, the electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized, wherein the handle is configured to be actuated to unlock the lock for unlocking the door from the cabinet when the handle is disengaged from the electronic lock, and wherein the electronic lock is configured to re-lock only when the handle is engaged with the electronic lock. In other aspects, the electronic key comprises an internal power source, and wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source.

In another embodiment, a security system for a server rack is provided and includes an electronic key and an electronic lock configured to be attached to the server rack. The electronic lock is configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock. Moreover, the electronic key comprises an internal power source, and wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source.

In another embodiment, a security system for securing items from theft is provided. The security system comprises an electronic key and an electronic lock configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock. The electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized, wherein the handle is configured to be actuated to unlock the lock for accessing the items when the handle is disengaged from the electronic lock, and wherein the electronic lock is configured to re-lock only when the handle is engaged with the electronic lock.

In another embodiment, a method for protecting a server rack from unauthorized access is provided. The method comprises attaching an electronic lock to a server rack and actuating an electronic key for communicating with the electronic lock to determine whether the electronic key is authorized to unlock or lock the electronic lock. The electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized. The method further includes actuating the handle to unlock the lock for accessing the server rack when the handle is disengaged from the electronic lock and actuating the handle to engage the electronic lock. In addition, the method includes actuating the electronic key to re-lock the electronic lock when the handle is engaged with the electronic lock.

In another embodiment, a method for protecting a server rack from unauthorized access is provided. The method comprises communicating between an electronic key and an electronic lock to determine whether the electronic key is authorized to unlock or lock the electronic lock. The method further includes transferring electrical power from the internal power source of the electronic key to the security device, wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source if the electronic key is authorized.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. **1**A shows an embodiment of a security system and method including a programmable electronic key, a security device, a programming station and a charging station according to an embodiment of the invention.

FIG. **1**B is an enlarged view showing the programmable electronic key of FIG. **1**A positioned on the programming station of FIG. **1**A to be programmed with a security code.

FIG. **2** further shows the system and method of FIG. **1**A with the programmable electronic key positioned to operate the security device.

FIG. **3**A further shows the system and method of FIG. **1**A with the programmable electronic key disposed on the charging station.

FIG. **3**B is an enlarged view showing the programmable electronic key of FIG. **1**A positioned on the charging station of FIG. **1**A to recharge a power source disposed within the key.

FIG. **4** is an enlarged view showing the security device of the system and method of FIG. **1**A.

FIG. **5** is an enlarged view showing the programmable electronic key of the system and method of FIG. **1**A in greater detail.

FIG. **6** is an exploded view of the programmable electronic key of FIG. **5**.

FIG. **7**A is a perspective view of the programmable electronic key of FIG. **5**.

FIG. **7**B is an end view of the programmable electronic key of FIG. **5**.

FIG. **8** is a perspective view showing a lengthwise cross-section of the programmable electronic key of FIG. **5**.

FIG. **9**A is a top view showing the charging station of the system and method of FIG. **1**A.

FIG. **9**B is a perspective view showing a diagonal cross-section of the charging station of FIG. **9**A taken along the line **9**B-**9**B.

FIG. **10** shows another embodiment of a security system and method including a programmable electronic key, a security device, a programming station and a charging station according to an embodiment of the invention.

FIG. **11** is an enlarged view showing the programmable electronic key of FIG. **10** positioned on the charging station of FIG. **10** to recharge a power source disposed within the key.

FIG. **12** is an enlarged view showing the security device of the system and method of FIG. **10**.

FIG. **13** is an enlarged view showing the programmable electronic key of the system and method of FIG. **10** in greater detail.

FIG. **14** is a perspective view showing a pair of matched coils for use with the programmable electronic key and the security device of FIG. **10**.

FIG. **15**A is a perspective view of the programmable electronic key of FIG. **13**.

FIG. **15**B is an end view of the programmable electronic key of FIG. **13**.

FIG. **16** is a perspective view showing a lengthwise cross-section of the programmable electronic key of FIG. **13**.

FIG. **17**A is a top view showing the charging station of the system and method of FIG. **10**.

FIG. **17**B is a perspective view showing a diagonal cross-section of the charging station of FIG. **17**A taken along the line **17**B-**17**B.

FIG. **18** illustrates a system comprising a server rack and a lock according to an embodiment of the invention.

FIG. **19** is rear perspective view of the server rack and the lock of FIG. **19**.

FIGS. **20** and **21** illustrate a partial perspective view of an electronic lock in a locked state and an unlocked state according to an embodiment of the invention.

FIGS. **22** and **23** illustrate an electronic lock and a handle in a disengaged position and an engaged position, respectively, according to an embodiment of the invention.

FIGS. **24-27** illustrate side cross-sectional views of an electronic lock with the handle being in a disengaged position and an engaged position according to embodiments of the invention.

FIG. **28** illustrates a partial perspective view of an electronic lock according to an embodiment of the invention.

FIG. **29** illustrates a side cross-sectional view of the electronic lock shown in FIG. **28** with the handle in an engaged position.

FIG. **30** illustrates a partial perspective view of an electronic lock in a locked state according to an embodiment of the invention.

FIG. **31** illustrates the lock of FIG. **30** in an unlocked state.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

Referring now to the accompanying drawing figures wherein like reference numerals denote like elements throughout the various views, one or more embodiments of a security system and method for server racks are shown. In the embodiments shown and described herein, the system and method include a programmable electronic key, indicated generally at **20**, **120** and a security device, indicated generally at **40**, **140**, **240**, **340**, **440**. Security devices **40**, **140**, **240**, **340**, **440** suitable for use with the programmable electronic keys **20**, **120** include, but are not limited to, server racks for storing various types and quantities of computer and/or network equipment, such as for example, servers, computers, hard drives, media storage, routers, hubs, network switches, etc. The server rack may define an enclosure that is configured to secure various computer and/or network equipment that is only configured to be accessed by authorized personnel, such as described in the following embodiments. Of course, embodiments of the present invention are applicable to any number of security devices **40**, **140**, **240**, **340**, **440** for securing various items from theft and are therefore not intended to be limited to use with server racks or server cabinets.

An embodiment of a system and method according to the invention is illustrated in FIGS. **1**A-**9**B. The embodiment of the security system and method depicted comprises a programmable electronic key **20**, which is also referred to herein as a security key, and a security device **40** that is configured to be operated by the key. The system and method may further comprise an optional programming or authorization station, indicated generally at **60**, that is operable for programming the key **20** with a security code, which is also referred to herein as a Security Disarm Code (SDC). The term SDC is not intended to be limiting, as it may be any code configured to be used to determine whether the key **20** is authorized to control the security device **40**. In addition to programming station **60**, the system and method may further comprise an optional charging station, indicated generally at **80**, that is operable for initially charging and/or subsequently recharging a power source disposed within the key **20**. For example, security key **20** and security device **40** may each be programmed with the same SDC into a respective permanent memory. The security key **20** may be provisioned with a single-use (e.g., non-rechargeable) power source, such as a conventional or extended-life battery, or alternatively, the key may be provisioned with a multiple-use (e.g., rechargeable) power source, such as a conventional capacitor or rechargeable battery. In either instance, the power source may be permanent, semi-permanent (e.g., replaceable), or rechargeable, as desired. In the latter instance, charging station **80** is provided to initially charge and/or to subsequently recharge the power source provided within the security key **20**. Furthermore, key **20** and/or security device **40** may be provided with only a transient memory, such that the SDC must be programmed (or reprogrammed) at predetermined time intervals. In this instance, programming station **60** is provided to initially program and/or to subsequently reprogram the SDC into the key **20**. As will be described, key **20** is operable to initially program and/or to subsequently reprogram the security device **40** with the SDC. Key **20** is then further operable to operate the security device **40** using power transferred to the security device and/or data communicated with the device, as will be described.

In one embodiment of the system and method illustrated in FIGS. **1**A-**9**B, programmable electronic key **20** is configured to be programmed with a unique SDC by the programming station **60**. A programming station **60** suitable for use with the present invention is shown and described in detail in the commonly owned U.S. Pat. No. 7,737,844 entitled PROGRAMMING STATION FOR A SECURITY SYSTEM FOR PROTECTING MERCHANDISE, the disclosure of which is incorporated herein by reference in its entirety. As illustrated in FIG. **1**A and best shown in enlarged FIG. **1**B, the key **20** is presented to the programming station **60** and communication therebetween is initiated, for example by pressing a control button **22** provided on the exterior of the key. Communication between the programming station **60** and the key may be accomplished directly, for example, by one or more electrical contacts, or indirectly, for example by wireless communication. Any form of wireless communication capable of transferring data between the programming station **60** and key **20** is also possible, including without limitation optical transmission, acoustic transmission, or magnetic induction. In the, embodiments shown and described herein, communication between programming station **60** and key **20** is accomplished by wireless optical transmission, and more particularly, by cooperating infrared (IR) transceivers provided in the programming station and the key. The components and method of IR communication between programming station **60** and key **20** is described in greater detail in the aforementioned U.S. Pat. No. 7,737,844, and accordingly, will not be repeated here. For the purpose of describing the present invention, it is sufficient that the programming station comprises at least a logic control circuit for generating or being provided with a SDC, a memory for storing the SDC, and a communications system suitable for interacting with the programmable electronic key **20** in the manner described herein to program the key with the SDC.

As shown in FIG. **1**B, programming station **60** comprises a housing **61** configured to contain the logic control circuit that generates the SDC, the memory that stores the SDC, and a communications system, namely an optical transceiver, for wirelessly communicating the SDC to a cooperating optical transceiver disposed within the key **20**. In use, the logic control circuit generates the SDC, which may be a predetermined (e.g., "factory preset") security code, or which may be a security code that is randomly generated by the logic control circuit of the programming station **60** at the time a first key **20** is presented to the station for programming In the latter instance, the logic control circuit further comprises a random number generator for producing the unique SDC. A series of visual indicators, for example light-emitting diodes (LEDs) **67** may be provided on the exterior of the housing **61** for indicating the operating status of the programming station. Use of the programming station **60** may further require authorization, such as with a mechanical lock mechanism, for example, a conventional key and tumbler lock **68**, for preventing use of the programming station by an unauthorized person. Alternatively, the programming station **60** may require various other forms of authentication, such as a pin code, biometric identification, facial recognition, etc. in order to activate the key **20** or otherwise gain access to the key. As shown herein, the programming station **60** may be operatively connected to an external power source by a power cord **70** having at least one conductor. Alternatively, the programming station **60** may comprise an internal power source, for example an extended-life replaceable battery or a rechargeable battery, for providing power to the logic control circuit and the LEDs **67**.

In one example embodiment, the logic control circuit of the programming station **60** performs an electronic exchange of data with a logic control circuit of the key **20**, commonly referred to as a "handshake communication protocol." The handshake communication protocol determines whether the key is an authorized key that has not been programmed previously, or is an authorized key that is being presented to the programming station a subsequent time to refresh the SDC. In the event that the handshake communication protocol fails, the programming station **60** will not provide the SDC to the unauthorized device attempting to obtain the SDC, for example an infrared reader on a counterfeit key. When the handshake communication protocol succeeds, programming station **60** permits the SDC randomly generated by the logic control circuit and/or stored in the memory of the station to be transmitted by the optical transceiver to the cooperating optical transceiver disposed within the key **20**. As will be readily apparent to those skilled in the art, the SDC may be transmitted from the programming station **60** to the security key **20** alternatively by any other suitable means, including without limitation, electrical contacts or electromechanical, electromagnetic or magnetic conductors, as desired.

As illustrated in FIG. **2**, the security key **20** programmed with the SDC is then positioned to operatively engage the security device **40**. In the embodiments shown and described herein, the security device is a conventional cabinet lock that has been modified to be unlocked by the programmable electronic key **20**. Preferably, the security device **40** is a "passive" device. As used herein, the term passive is intended to mean that the security device **40** does not have an internal power source sufficient to lock and/or unlock a mechanical lock mechanism. Significant cost savings are obtained by a retailer when the security device **40** is passive since the expense of an internal power source is confined to the security key **20**, and one such key is able to operate multiple security devices. If desired, the security device **40** may also be provided with a temporary power source (e.g., capacitor or limited-life battery) having sufficient power to activate an alarm, for example a piezoelectric audible alarm, that is actuated by a sensor, for example a contact, proximity or limit switch, in response to a security breach. The temporary power source may also be sufficient to communicate data, for example a SDC, from the security device **40** to the security key **20** to authenticate the security device and thereby authorize the key to provide power to the security device. With this embodiment of the present invention, the mechanical lock mechanism is operated by electrical power that is transferred from the key **20** to the security device **40** via electrical contacts, as will be described.

The security device **40** further comprises a logic control circuit, similar to the logic control circuit disposed within the key **20**, adapted to perform a handshake communication protocol with the logic control circuit of the key in essentially the same manner as that between the programming station **60** and the key. In essence, the logic control circuit of the key **20** and the logic control circuit of the security device **40** communicate with each other to determine whether the security device is an authorized device that does not have a security code, or is a device having a proper (e.g., matching) SDC. The key **20** may be configured to initially transfer power to the security device **40** in the event the security device is a passive device to allow the security device to communicate with the key. In the event the handshake communication protocol fails (e.g., the device is not authorized or the device has a non-matching SDC), the key **20** will not program the device **40** with the SDC, and consequently, the security device will not operate. If the security device **40** was previously programmed with a different SDC, the device will no longer communicate with the security key **20**. In the event the handshake communication protocol is successful, the security key **20** permits the SDC stored in the key to be transmitted by the optical transceiver disposed within the key to a cooperating optical transceiver disposed within the security device **40** to program the device with the SDC. As will be readily apparent to those skilled in the art, the SDC may be transmitted from the security key **20** to the security device **40** alternatively by any other suitable means, including without limitation, via one or more electrical contacts, or via electromechanical, electromagnetic or magnetic conductors, as desired. Furthermore, the SDC may be transmitted by inductive transfer of data from the programmable electronic key **20** to the programmable security device **40**.

On the other hand, when the handshake communication protocol is successful and the security device **40** is an authorized device having the same (e.g., matching) SDC, the mechanical lock mechanism of the security device **40** may operate using power from the key **20**, either power that had been previously transferred by the key and stored by the security device and/or by power transmitted by the key to the security device. In the embodiment of FIGS. **1**A-**9**B, electrical contacts disposed on the security key **20** electrically couple with cooperating electrical contacts on the security device **40** to transfer power from the internal battery of the key to the security device. Power may be transferred directly to the mechanical lock mechanism, or alternatively, may be transferred to a power circuit disposed within the security device **40** that operates the mechanical lock mechanism of the security device and may be configured to store the power for subsequent operation of the lock mechanism. In the embodiment of FIGS. **1**A-**9**B, the cabinet lock **40** is affixed to one of the pair of adjacent and overlapping sliding doors **102** of a conventional cabinet **100**. The cabinet **100** typically contains various types of equipment **110**. The doors **102** overlap medially between the ends of the cabinet **100** and the cabinet lock **40** is secured on an elongate locking arm **104** of a lock bracket **105** affixed to the inner door. In the illustrated example, the key **20** transfers power to an electric motor, such as a DC stepper motor, solenoid, or the like, that unlocks the lock mechanism of the cabinet lock **40** so that the cabinet lock can be removed from the arm **104** of the bracket **105** and the doors moved (e.g., slid) relative to one another to access the equipment **110** stored within the cabinet **100**. As shown, the arm **104** of the bracket **105** is provided with one-way ratchet teeth **106** and the cabinet lock **40** is provided with a complimentary ratchet pawls (not shown) in a conventional manner so that the key **20** is not required to lock the cabinet lock **40** onto the inner door **102** of the cabinet **100**. If desired, however, the cabinet lock **40** can be configured to require use of the key **20** to both unlock and lock the cabinet lock.

It will be readily apparent to those skilled in the art that the cabinet lock illustrated herein is but one of numerous types of passive security devices **40** that can be configured to be operated by a programmable electronic key **20** according to the present invention. In any of the aforementioned embodiments, the security device **40** may further comprise an electronic lock mechanism, such as a conventional proximity, limit or contact switch, including an associated monitoring circuit that activates an alarm in response to the switch being actuated or the integrity of a sense loop monitored by the monitoring circuit being compromised. In such embodiments the security device **40** comprises a logic control circuit, or the equivalent, including a memory for storing a SDC, and a communication system for initially receiving the SDC from the security key **20** and subsequently communicating with the key to authenticate the SDC of the key.

As illustrated in FIG. **3**A and shown enlarged in FIG. **3**B, the security system and method further comprises charging station **80** for initially charging and subsequently recharging a rechargeable battery disposed within the security key **20**. The charging station **80** comprises at least one charging port **82** sized and shaped to receive a key **20** to be charged or recharged. As will be described in greater detail with reference to FIGS. **9**A and **9**B, each charging port **82** comprises at least one magnet **85** for securely positioning and retaining the key **20** within the charging port **82** in electrical contact with the charging station **80**. If desired, the charging station **80** may comprise an internal power source, for example, an extended-life replaceable battery or a rechargeable battery, for providing power to up to four keys **20** positioned within respective charging ports **82**. Alternatively, and as shown herein, charging station **80** may be operatively connected to an external power source by a power cord **90** having at least one conductor.

An available feature of a security system and method according to the invention is that the logic control circuit of the programmable electronic key **20** may include a time-out function. More particularly, the ability of the key **20** to transfer data and power to the security device **40** is deactivated after a predetermined time period. By way of example, the logic control circuit may be deactivated after about eight hours from the time the key was programmed or last refreshed by the programming station **60**. Thus, an authorized sales associate typically must program or refresh the key **20** assigned to him at the beginning of each work shift. Furthermore, the charging station **80** may be configured to deactivate the logic control circuit of the key **20** (and thereby prevent use of the SDC) when the key is positioned within a charging port **82**. In this manner, the charging station **80** can be made available to an authorized sales associate in an unsecured location without risk that a charged key **20** could be removed from the charging station and used to maliciously disarm and/or unlock a security device **40**. The security key **20** would then have to be programmed or refreshed with the SDC by the programming station **60**, which is typically monitored or maintained at a secure location, in order to reactivate the logic control circuit of the key. If desired, the charging station **80** may alternatively require a matching handshake communication protocol with the programmable electronic key **20** in the same manner as the security device **40** and the key.

FIG. **4** is an enlarged view showing the embodiment of the security device **40** in greater detail. As previously mentioned, a security device **40** according to the present invention may utilize electrical power to lock and/or unlock a mechanical lock mechanism, and optionally, further includes an electronic lock mechanism, such as an alarm or a security "handshake." At the same time, the security device **40** must be a passive device in the sense that it does not have an internal power source sufficient to operate the mechanical lock mechanism. As a result, the security device **40** must be configured to receive at least power, and preferably, both power and data from an external source, such as the security key **20** shown and described herein. The embodiment of the security device depicted in FIG. **4** is a cabinet lock **40** configured to be securely affixed to the locking arm **104** of a conventional cabinet lock bracket **105**, as previously described. The cabinet lock **40** comprises a logic control circuit for performing a security handshake communication protocol with the logic control circuit of the security key **20** and for being programmed with the SDC by the key. In other embodiments, the cabinet lock **40** may be configured to transmit the SDC to the security key **20** to authenticate the security device and thereby authorize the key to transfer power to the cabinet lock. As previously mentioned, the data (e.g., handshake communication protocol and SDC) may be transferred (e.g., transmitted and received) by electrical contacts, optical transmission, acoustic transmission or magnetic induction, for example.

The cabinet lock **40** comprises a housing **41** sized and shaped to contain a logic control circuit (not shown) and an internal mechanical lock mechanism (not shown). A transfer port **42** formed in the housing **41** is sized and shaped to receive a transfer probe of the security key **20**, as will be described. At least one magnet **45** is disposed within the transfer port **42** for securely positioning and retaining the transfer probe of the key **20** in electrical contact with electrical contacts of the mechanical lock mechanism, and if desired, in electrical contact with the logic control circuit of the cabinet lock **40**. In the embodiment shown and described in FIGS. **1**A-**9**B, data is transferred from the security key **20** to the cabinet lock **40** by wireless communication, such as by infrared (IR) optical transmission, as shown and described in the commonly owned U.S. Pat. No. 7,737,843 entitled PROGRAMMABLE ALARM MODULE AND SYSTEM FOR PROTECTING MERCHANDISE, the disclosure of which is incorporated herein by reference in its entirety. Power is transferred from the security key **20** to the cabinet lock **40** through electrical contacts disposed on the transfer probe of the key and corresponding electrical contacts disposed within the transfer port **42** of the cabinet lock. For example, the transfer port **42** may comprise a metallic outer ring **46** that forms one electrical contact, while at least one of the magnets **45** form another electrical contact to complete an electrical circuit with the electrical contacts disposed on the transfer probe of the key **20**. Regardless, electrical contacts transfer power from the key **20** to the mechanical lock mechanism disposed within the housing **41**. As previously mentioned, the power transferred from the key **20** is used to operate the mechanical lock mechanism, for example utilizing an electric motor, DC stepper motor, solenoid, or the like, to unlock the mechanism so that the cabinet lock **40** can be removed from the locking arm **104** of the lock bracket **105**.

FIGS. **5-8** show an embodiment of a security key, also referred to herein as a programmable electronic key, **20** according to the present invention. As previously mentioned, the security key **20** is configured to transfer both data and power to a security device **40** that comprises an electronic lock mechanism and a mechanical lock mechanism, as previously described. Accordingly, the programmable electronic key **20** must be an "active" device in the sense that it has an internal power source sufficient to operate the mechanical lock mechanism of the security device **40**. As a result, the programmable electronic key **20** may be configured to transfer both data and power from an internal source disposed within the key, for example a logic control circuit and a battery. The embodiment of the programmable electronic key **20** depicted in FIGS. **5-8** is a security key configured to be received within the transfer port **42** of the cabinet lock **40** shown in FIG. **4**, as well as within the programming port **62** of the programming station **60** (FIG. **2**; FIG. **3**A) and the charging port **82** of the charging station **80** (FIG. **3**B; FIG. **9**A; FIG. **9**B). The programmable electronic key **20** comprises a logic control circuit for performing a handshake communication protocol with the logic control circuit of the programming station **60** and for receiving the SDC from the programming station, as previously described. The logic control circuit of the programmable electronic key **20** further performs a handshake communication protocol with the logic control circuit of the security device **40** and transfers the SDC to the device or permits operation of the device, as previously described. As previously mentioned, the data (e.g., handshake communication protocol and SDC) may be transferred by direct electrical contacts, optical transmission, acoustic transmission or magnetic induction.

As illustrated in FIG. **6**, the programmable electronic key **20** comprises a housing **21** and an outer sleeve **23** that is removably disposed on the housing. The housing **21** contains the internal components of the key **20**, including without limitation the logic control circuit, memory, communication system and battery, as will be described. A window **24** may be formed through the outer sleeve **23** for viewing indicia **24**A that uniquely identifies the key **20**, or alternatively, indicates a particular server rack for use with the key. The outer sleeve **23** is removably disposed on the housing **21** so that the indicia **24**A may be altered or removed and replaced with different indicia. The programmable electronic key **20** may further comprise a detachable "quick-release" type key chain ring **30**. An opening **26** (FIG. **8**) is formed through the outer sleeve **23** and a key chain ring port **28** is formed in the housing **21** for receiving the key chain ring **30**. The programmable electronic key **20** further comprises a transfer probe **25** located at an end of the housing **21** opposite the key chain ring port **28** for transferring data and power to the security device **40**, as previously described. The transfer probe **25** also transmits and receives the handshake communication protocol and the SDC from the programming station **60**, as previously described, and receives power from the charging station **80**, as will be described in greater detail with reference to FIG. **9**A and FIG. **9**B.

As best shown in FIG. **8**, an internal battery **31** and a logic control circuit, or printed circuit board (PCB) **32** are disposed within the housing **21** of the programmable electronic key **20**. Battery **31** may be a conventional extended-life replaceable battery, but preferably, is a rechargeable battery suitable for use with the charging station **80**. The logic control circuit **32** is operatively coupled and electrically connected to a switch **33** that is actuated by the control button **22** provided on the exterior of the key **20** through the outer sleeve **23**. Control button **22** in conjunction with switch **33** controls certain operations of the logic control circuit **32**, and in particular, transmission of the data to the security device **40**. In that regard, the logic control circuit **32** is further operatively coupled and electrically connected to a communication system **34** for transmitting and receiving the handshake communication protocol and SDC data. In the embodiment shown and described herein, the communication system **34** is a wireless infrared (IR) transceiver for optical transmission of data between the programmable electronic key **20** and the programming station **60**, as well as between the key **20** and the security device **40**. As a result, the transfer probe **25** of the key **20** is provided with an optically transparent or translucent filter window **35** for emitting and collecting optical transmissions between the key **20** and the programming station **60**, or alternatively, between the key **20** and the security device **40**, as required. Transfer probe **25** further comprises a pair of bi-directional power transfer electrical contacts **36**, **38** made of an electrically conductive material for transferring power to the security device **40** and for receiving power from the charging station **80**, as required. Accordingly, electrical contacts **36**, **38** are electrically connected to battery **31**, and are operatively coupled and electrically connected to logic control circuit **32** in any suitable manner, for example by conductive insulated wires or plated conductors.

An important aspect of a programmable electronic key **20** according to the present invention, especially when used for use in conjunction with a security device **40** as described herein, is that the key does not require a physical force to be exerted by a user on the key to operate the mechanical lock mechanism of the security device. By extension, no physical force is exerted by the key on the mechanical lock mechanism. As a result, the key cannot be unintentionally broken off in the lock, as often occurs with conventional mechanical key and lock mechanisms. Furthermore, neither the key nor and the mechanical lock mechanism suffer from excessive wear as likewise often occurs with conventional mechanical key and lock mechanisms. In addition, there is no required orientation of the transfer probe **25** of the programmable electronic key **20** relative to the charging port **82** of the charging station **80** or the transfer port **42** of the security device **40**. Accordingly, any wear of the electrical contacts on the transfer probe **25**, the charging port **82** or the transfer port **42** is minimized. As a further advantage, an authorized person is not required to position the transfer probe **25** of the programmable electronic key **20** in a particular orientation relative to the transfer port **42** of the security device **40** and thereafter exert a compressive and/or torsional force on the key to operate the mechanical lock mechanism of the device.

FIG. **9**A and FIG. **9**B show charging station **80** in greater detail. As previously mentioned, the charging station **80** recharges the internal battery **31** of the programmable electronic key **20**, and if desired, deactivates the data transfer and/or power transfer capability of the key until the key is reprogrammed with the SDC by the programming station **60**. Regardless, the charging station **80** comprises a housing **81** for containing the internal components of the charging station. The exterior of the housing **81** has at least one, and preferably, a plurality of charging ports **82** formed therein that are sized and shaped to receive the transfer probe **25** of the security key **20**, as previously described. At least one magnet **85** is disposed within each charging port **82** for securely positioning and retaining the transfer probe **25** in electrical contact with the charging station **80**. More particularly, the electrical contacts **36**, **38** of the key **20** are retained within the charging port **82** in electrical contact with the magnets **85** and a resilient "pogo" pin **86** made of a conductive material to complete an electrical circuit between the charging station **80** and the battery **31** of the key.

As best shown in FIG. **9**B, housing **81** is sized and shaped to contain a logic control circuit, or printed circuit board (PCB) **92** that is operatively coupled and electrically connected to the magnets **85** and the pogo pin **86** of each charging port **82**. The pogo pin **86** is depressible to complete an electrical circuit as the magnets **85** position and retain the electrical contacts **36**, **38** within the charging port **82**. In particular, magnets **85** make electrical contact with the outer ring electrical contact **36** of the transfer probe **25** of key **20**, while pogo pin **86** makes electrical contact with inner ring electrical contact **38** of the transfer probe. When the pogo pin **86** is depressed and the electrical circuit between the charging station **80** and the key **20** is completed, the charging station recharges the internal battery **31** of the key. As previously mentioned, charging station **80** may comprise an internal power source, for example, an extended-life replaceable battery or a rechargeable battery, for providing power to the key(s) **20** positioned within the charging port(s) **82**. Alternatively, and as shown herein, the logic control circuit **92** of the charging station **80** is electrically connected to an external power source by a power cord **90** having at least one conductor. Furthermore, logic control circuit **92** may be operable for deactivating the data transfer and power transfer functions of the programmable electronic key **20**, or alternatively, for activating the "time-out" feature of the key until it is reprogrammed or refreshed by the programming station **60**.

FIGS. **10**-**17**B show another embodiment of a security system and method including a programmable key, a security device, a programming station, and a charging station according to various embodiments of the present invention. In this embodiment, the system and method comprise at least a programmable electronic key (also referred to herein as a security key) with inductive transfer, indicated generally at **120**, and a security device with inductive transfer capability, indicated generally at **140**, that is operated by the key **120**. The programmable electronic key **120** is useable with any security device or locking device, such as various types of server racks as discussed above, with inductive transfer capability that requires power transferred from the key to the device by induction, or alternatively, requires data transferred between the key and the device and power transferred from the key to the device by induction.

As illustrated in FIG. **11**, the security system and method may further comprise a charging station **180** for initially charging and subsequently recharging a rechargeable battery disposed within the security key **120** via inductive transfer. The charging station **180** comprises at least one charging port **182** sized and shaped to receive a security key **120**. If desired, each charging port **182** may comprise mechanical or magnetic means for properly positioning and securely retaining the key **120** within the charging port. By way of example and without limitation, at least one, and preferably, a plurality of magnets (not shown) may be provided for positioning and retaining the key **120** within the charging port **182** of the charging station **180**. However, as will be described further with reference to FIG. **17**B, it is only necessary that the inductive transceiver of the security key **120** is sufficiently aligned with the corresponding inductive transceiver of the charging station **180** over a generally planar surface within the charging port **182**. Thus, magnets are not required (as with charging station **80**) to position, retain and maintain electrical contacts provided on the security key **120** in electrical contact with corresponding electrical contacts provided on the charging station **180**. If desired, the charging station **180** may comprise an internal power source, for example, an extended-life replaceable battery or a rechargeable battery, for providing power to the key(s) **120** positioned within the charging port(s) **182**. Alternatively, and as shown herein, charging station **180** may be operatively connected to an external power source by a power cord **190** having at least one conductor in a conventional manner.

FIG. **12** shows the security device **140** with inductive transfer in greater detail. In a particular embodiment, a security device **140** with inductive transfer according to the invention may both receive electrical power from the security key **120** and communicate (e.g., transmit/receive) the SDC with the key by magnetic induction.

The cabinet lock **140** comprises a housing **141** sized and shaped to contain a logic control circuit (not shown) and an internal mechanical lock mechanism (not shown). A transfer port **142** formed in the housing **141** is sized and shaped to receive a transfer probe of the security key **120**, as will be described. If desired, the transfer port **142** may comprise mechanical or magnetic means for properly positioning and securely retaining the key **120** within the transfer port. By way of example and without limitation, at least one, and preferably, a plurality of magnets (not shown) may be provided for positioning and retaining the key **120** within the transfer port **142** of the cabinet lock **140**. However, as previously described with respect to the security key **120** and the charging port **182** of the charging station **180**, it is only necessary that the inductive transceiver of the security key **120** is sufficiently aligned with the corresponding inductive transceiver of the cabinet lock **140** over a generally planar surface within the transfer port **42**. Therefore, magnets are not required to position, retain and maintain electrical contacts provided on the security key **120** in electrical contact with corresponding electrical contacts provided on the cabinet lock **140**. In the particular embodiment shown and described herein, data is transferred from the security key **120** to the cabinet lock **140** by wireless communication, such as infrared (IR) optical transmission as shown and described in the aforementioned U.S. Pat. No. 7,737,843. Power is transferred from the security key **120** to the cabinet lock **140** by induction across the transfer port **142** of the cabinet lock using an inductive transceiver disposed within a transfer probe of the key that is aligned with a corresponding inductive transceiver disposed within the cabinet lock. For example, the transfer probe of the security key **120** may comprise an inductive transceiver coil that is electrically connected to the logic control circuit of the key to provide electrical power from the internal battery of the key to an inductive transceiver coil disposed within the cabinet lock **140**. The inductive transceiver coil of the cabinet lock **140** then transfers the electrical power from the internal battery of the key **120** to the mechanical lock mechanism disposed within the housing **141** of the cabinet lock. As previously mentioned, the power transferred from the key **120** is used to unlock the mechanical lock mechanism, for example utilizing an electric motor, DC stepper motor, solenoid, or the like, so that the cabinet lock **140** can be removed from the arm **104** of the lock bracket **105**.

FIGS. **13-16** show the programmable electronic key **120** with inductive transfer in greater detail. As previously mentioned, the key **120** is configured to transfer both data and power to a security device **140** that comprises an electronic lock mechanism and a mechanical lock mechanism. Accordingly, the programmable electronic key **120** must be an active device in the sense that it has an internal power source sufficient to operate the mechanical lock mechanism of the security device **140**. As a result, the programmable electronic key **120** may be configured to transfer both data and power from an internal source, such as a logic control circuit and a battery disposed within the key. The embodiment of the programmable electronic key **120** depicted herein is a security key with inductive transfer capability configured to be received within the transfer port **145** of the cabinet lock **140** shown in FIG. **12**, as well as the programming port **62** of the programming station **60** (FIG. **2**) and the charging port **182** of the charging station **180** (FIG. **11**). The programmable electronic key **120** comprises a logic control circuit for performing a handshake communication protocol with the logic control circuit of the programming station **60** and for receiving the SDC from the programming station, as previously described. The logic control circuit of the programmable electronic key **120** further performs a handshake communication protocol with the logic control circuit of the security device **140** and transfers the SDC to the security device, as previously described. In a particular embodiment, a security key **120** with inductive transfer according to the invention may both transfer electrical power to a security device **140** and communicate the SDC with the security device by magnetic induction.

The programmable electronic key **120** comprises a housing **121** having an internal cavity or compartment that contains the internal components of the key, including without limitation the logic control circuit, memory, communication system and battery, as will be described. As shown, the housing **121** is formed by a lower portion **123** and an upper portion **124** that are joined together after assembly, for example by ultrasonic welding. The programmable electronic key **120** further defines an opening **128** at one end for coupling the key to a key chain ring, lanyard or the like. As previously mentioned, the programmable electronic key **120** further comprises a transfer probe **125** located at an end of the housing **121** opposite the opening **128** for transferring data and power to the security device **140**. The transfer probe **125** is also operable to transmit and receive the handshake communication protocol and the SDC from the programming station **60**, as previously described, and to receive power from the charging station **180**, as will be described in greater detail with reference to FIG. **17**A and FIG. **17**B.

FIG. **14** shows an embodiment of an inductive coil **126** having high magnetic permeability that is adapted to be disposed within the housing **121** of the electronic key **120** adjacent the transfer probe **125**. As shown herein, the inductive coil **126** comprises a highly magnetically permeable ferrite core **127** surrounded by a plurality of inductive core windings **129**. The inductive core windings **129** consist of a length of a conductive wire that is wrapped around the ferrite core. As is well known, passing an alternating current through the conductive wire generates, or induces, a magnetic field around the inductive core **127**. The alternating current in the inductive core windings **129** may be produced by connecting the leads **129**A and **129**B of the conductive wire to the internal battery of the electronic key **120** through the logic control circuit. FIG. **14** further shows an inductive coil **146** having high magnetic permeability that is adapted to be disposed within the housing **141** of the security device (e.g., cabinet lock) **140** adjacent the transfer port **142**. As shown herein, the inductive coil **146** comprises a highly magnetically permeable ferrite core **147** surrounded by a plurality of inductive core windings **149** consisting of a length of a conductive wire that is wrapped around the ferrite core. Placing the transfer probe **125** of the electronic key **120** into the transfer port **142** of the cabinet lock **140** and passing an alternating current through the inductive core windings **129** of the inductive core **126** generates a magnetic field within the transfer port of the cabinet lock in the vicinity of the inductive coil **146**. As a result, an alternating current is generated, or induced, in the conductive wire of the inductive core windings **149** of inductive coil **146** having leads **149**A and **149**B connected to the logic control circuit of the cabinet lock **140**. The alternating current induced in the inductive coil **146** of the cabinet lock **140** is then transformed into a direct current in a known manner, such as via a bridge rectifier on the logic control circuit, to provide direct current (DC) power to the cabinet lock. The DC power generated in the cabinet lock **140** by the inductive coil **126** of the electronic key **120**, may be used, for example, to unlock a mechanical lock mechanism disposed within the housing **141** of the cabinet lock.

As best shown in FIG. **16**, an internal battery **131** and a logic control circuit, or printed circuit board (PCB) **132** are disposed within the housing **121** of the programmable electronic key **120**. Battery **131** may be a conventional extended-life replaceable battery, but preferably, is a rechargeable battery suitable for use with the charging station **180**. The logic control circuit **132** is operatively coupled and electrically connected to a switch **133** that is actuated by the control button **122** provided on the exterior of the key **120** through the housing **121**. Control button **122** in conjunction with switch **133** controls certain operations of the logic control circuit **132**, and in particular, transmission of the data (e.g., handshake communication protocol and SDC) between the key and the programming station **60**, as well as between the key and the security device **140**. In that regard, the logic control circuit **132** is further operatively coupled and electrically connected to a communication system **134** for transferring the handshake communication protocol and SDC data. As shown and described herein, the communication system **134** is a wireless infrared (IR) transceiver for optical transmission of data between the programmable electronic key **120** and the programming station **60**, and between the key and the security device **140**. As a result, the transfer probe **125** of the key **120** is provided with an optically transparent or translucent filter window **135** for emitting and collecting optical transmissions between the key **120** and the programming station **60**, or between the key and the security device **140**, as required. Transfer probe **125** further comprises inductive coil **126** (FIG. **14**) comprising inductive core **127** and inductive core windings **129** for transferring electrical power to the security device **140** and/or receiving electrical power from the charging station **180** to charge the internal battery **131**, as required. Accordingly, the leads **129**A and **129**B (FIG. **14**) of the inductive coil **126** are electrically connected to the logic control circuit **132**, which in turn is electrically connected to the battery **131**, in a suitable manner, for example by conductive insulated wires or plated conductors. Alternatively, the optical transceiver **134** may be eliminated and data transferred between the programmable electronic key **120** and the security device **140** via magnetic induction through the inductive coil **126**.

As noted above, one aspect of a programmable electronic key **120** according to the present invention, especially when used for use in conjunction with a security device **140** as described herein, is that the key does not require a physical force to be exerted by a user on the key to operate the mechanical lock mechanism of the security device. In addition, there is no required orientation of the transfer probe **125** of the programmable electronic key **120** relative to the charging port **182** of the charging station **180** or the transfer port **142** of the security device **140**. Accordingly, any wear of the electrical contacts on the transfer probe **125**, the charging port **182** or the transfer port **142** is minimized. As a further advantage, an authorized person is not required to position the transfer probe **125** of the programmable electronic key **120** in a particular orientation relative to the transfer port **142** of the security device **140** and thereafter exert a compressive and/or torsional force on the key to operate the mechanical lock mechanism of the device.

FIG. **17**A and FIG. **17**B show charging station **180** with inductive transfer capability in greater detail. As previously mentioned, the charging station **180** recharges the internal battery **131** of the security key **120**. In certain instances, the charging station **180** also deactivates the data transfer and/or power transfer capability of the key **120** until the key has been reprogrammed with the SDC by the programming station **60**. Regardless, the charging station **180** comprises a housing **181** for containing the internal components of the charging station. The exterior of the housing **181** has at least one charging port **182** formed therein that are sized and shaped to receive the transfer probe **125** of a programmable electronic key **120**. As previously described, mechanical or magnetic means may be provided for properly positioning and securely retaining the transfer probe **125** within the charging port **182** such that the inductive coil **126** is in alignment with a corresponding inductive coil **186** (FIG. **17**B) disposed within the housing **181** of the charging station **180** adjacent the charging port. As will be readily understood and appreciated, the inductive coil **186** adjacent the charging port **182** of the charging station **180** generates, or induces, an alternating current in the conductive wire of the inductive core windings **129** of inductive coil **126** that in turn provides DC power (for example, via a bridge rectifier on the logic control circuit **132**) to charge the battery **131** of the programmable electronic key **120**.

As best shown in FIG. **17**B, housing **181** is sized and shaped to contain a logic control circuit, or printed circuit board (PCB) **192** that is electrically connected and operatively coupled to an inductive coil **186** adjacent each of the charging ports **182**. In the manner previously described with respect to inductive coil **126** and inductive coil **146**, each inductive coil **186** comprises an inductive core **187** surrounded by a plurality of inductive core windings **189** formed by a conductive wire having a pair of leads (not shown). When an alternating current is passed through the conductive wire of the inductive core windings **189** with the transfer probe **125** of the programmable electronic key **120** disposed in the charging port **182** of the charging station **180**, the inductive coil **186** generates a magnetic field that induces an alternating current in the conductive wire of the inductive core windings **129** of the inductive coil **126** of the key. The alternating current in the inductive coil **126** is then transformed into DC power to charge the internal battery **131** of the programmable electronic key **120**. As previously mentioned, charging station **180** may comprise an internal power source, for example, an extended-life replaceable battery or a rechargeable battery, for providing power to the key(s) **120** positioned within the charging port(s) **182**. Alternatively, and as shown herein, the logic control circuit **192** of the charging station **180** is electrically connected to an external power source by a power cord **190** having at least one conductor. Furthermore, logic control circuit **192** may be operable for deactivating the data transfer and/or power transfer functions of the programmable electronic key **120**, or alternatively, for activating the "timing out" feature of the key until it is reprogrammed or refreshed by the programming station **60**.

In some embodiments, each electronic key **20**, **120** is configured to store various types of data. For example, each key **20**, **120** may store a serial number of one or more security devices **40**, **140**, **240**, **340**, **440**, the data and time of activation of the key, a user of the key, a serial number of the key, number of key activations, a type of activation (e.g., "naked" activation, activation transferring only data, activation transferring power, activation transferring data and power), and/or various events (e.g., a security device has been locked or unlocked). This information may be transmitted to a remote location or device (e.g., a backend computer) upon each activation of the key **20**, **120** or at any other desired period of time, such as upon communication with a programming station **60**. Thus, the data transfer may occur in predetermined time intervals or in real time or automatically in some embodiments. In some cases, the programming station **60** may be configured to store the data and transfer the data to a remote location or device. Authorized personnel may use this data to take various actions, such as to audit and monitor key user activity, audit security devices **40**, **140**, **240**, **340**, **440** (e.g., ensure the security devices are locked), etc. Moreover, such information may be requested and obtained on demand, such as from the programming station **60** and/or a remote device.

In other embodiments, the electronic key **20**, **120** is configured to obtain data from a security device **40**, **140**, **240**, **340**, **440**. For example, the security device **40**, **140**, **240**, **340**, **440** may store various data regarding past communication with a electronic key **20**, **120** (e.g., key identification, time of communication, etc.), and when a subsequent electronic key communicates with the same security device, the data is transferred to the electronic key. Thus, the security device **40**, **140**, **240**, **340**, **440** may include a memory for storing such data. In some cases, the security device **40**, **140**, **240**, **340**, **440** includes a power source for receiving and storing the data, while in other cases, the power provided by the electronic key **20**, **120** is used for allowing the merchandise security device to store the data. The electronic key **20**, **120** may then communicate the data for collection and review, such as at a remote location or device. In some instances, communication between the electronic key **20**, **120** and the programming station **60** may allow data to be pulled from the electronic key and communicated, such as to a remote location or device. In other cases, the electronic key **20**, **120** may be configured to obtain data from security devices **40**, **140**, **240**, **340**, **440**, such as an identification of the security device, identification of the items contained within or by the security device, and/or the system health of the security device and/or the items. The electronic key **20**, **120** may store the data and provide the data directly to a remote location or device or upon communication with the programming station **60**. As such, the electronic keys **20**, **120** may be a useful resource for obtaining various types of data from the merchandise security devices **40**, **140**, **240**, **340**, **440** without the need for wired connections or complex wireless networks or systems. In other embodiments, the security devices **40**, **140**, **240**, **340**, **440** themselves may include wireless communication capability to allow for transmission of the data to a remote device or location.

In another embodiment, each electronic key **20**, **120** may include a security code and a serial number for one or more security devices **40**, **140**, **240**, **340**, **440**. For example, a key **20**, **120** may only be able to lock or unlock a security device **40**, **140**, **240**, **340**, **440** where the security codes and the serial numbers match one another. In one example, each serial number is unique to a security device **40**, **140**, **240**, **340**, **440** and could be programmed at the time of manufacture or by the retailer. Individual electronic keys **20**, **120** may then be assigned particular serial numbers for authorized security devices **40**, **140**, **240**, **340**, **440** (e.g., user 1 includes serial numbers 1, 2, 3; user 2 includes serial numbers 1, 4, 5). Each of the electronic keys **20**, **120** may be programmed with the same security code using a programming station **60**. In order to lock or unlock a merchandise security device **40**, **140**, **240**, **340**, **440**, the electronic key **20**, **120** may communicate with a particular security device and determine whether the security codes and the serial numbers match. If the codes match, the electronic key **20**, **120** then locks or unlocks the security device **40**, **140**, **240**, **340**, **440**.

According to another embodiment, FIG. **18** illustrates a system **200** comprising a server rack **202** and a lock **240**. In this example, the server rack **202** includes a cabinet **204** and a door **206** pivotably attached to the cabinet, although other types of server racks may be used. The lock **240** is configured to lock the door **206** to the cabinet **204** such that the door is incapable of being opened when the lock is locked but is able to be opened when the lock is unlocked. FIG. **19** illustrates that in this embodiment, the lock **240** includes a latch **208** that is configured to engage the cabinet **204** to prevent the door **206** from opening when locked. The latch **208** may be any suitable mechanism configured to move between an engaged position with the cabinet **204** and a disengaged position whereby the latch is no longer in engagement with the cabinet.

In some embodiments, the lock **240** is configured to operate according to the various embodiment discussed above for the security devices **40**, **140**. For example, the lock **240** may be an electronic lock configured to be controlled by a key **20**, **120** using power and/or data communication using various communication protocols. In the illustrated embodiment, the lock **240** may include a transfer port **242** that is configured to facilitate communication with a key **20**, **120** as disclosed above (see, e.g., FIG. **23**). In other embodiments, the lock **240** may be configured to be operated using a combination of electrical and mechanical interaction. For example, an electronic key **20**, **120** may be used to indicate whether the operator is authorized to unlock the lock **240** and perform a first unlock operation, and the operator may be required to perform a second mechanical operation to disengage the latch **208** to allow the door **206** to be opened. Thus, in some embodiments, a two-step unlocking operation is required to unlock the lock **240**. In some cases, the lock **240** includes a handle **210**, and the operator of the lock may be required to move a handle to the unlocked position to unlock the door, such as by rotating the handle in a clockwise or counter-clockwise direction (see, e.g., FIGS. **22-23**). It is understood that the use of the term "handle" is not intended to be limiting, as any suitable actuator may be used to allow a mechanical disengagement of the lock **240** to allow the door **206** to be opened.

FIGS. **20-21** illustrate an example embodiment of an electronic lock **240** that is configured to release the handle **210** for allowing an operator to unlock the lock (a portion of the electronic lock has been removed for purposes of illustration). The electronic lock **240** may include a housing **241** that houses a variety of components as disclosed herein. In this embodiment, the lock **240** includes a mechanism configured to covert rotational movement into linear movement for releasing the handle **210**. In this regard, the lock **240** may include a motor **212** that is configured to rotate an actuator **214** (e.g., a cam) that is in engagement with a pin **216**. In this example, the motor **212** and the pin **216** are arranged in-line with one another or along the same axis. Rotation of the actuator **214** causes the pin **216** to move between an engaged position with the handle **210** (e.g., FIG. **20**) and a disengaged position with the handle (e.g., FIG. **21**). The pin **216** may be spring loaded in some cases to facilitate engagement and disengagement with the handle **210** as the actuator **214** rotates. The motor **212** may be operated using power transferred from a key **20**, **120**, as described above, or could include its own power source in other embodiments. The lock **240** could also include a power storage device (e.g., one or more capacitors) for storing power transmitted by the key **20**, **120** for performing one or more functions, such as operation of the motor **212**.

It is understood that a variety of mechanisms may be used for the electronic lock **240** to facilitate engagement and disengagement of the handle **210**. For example, FIGS. **28** and **29** show an alternative embodiment of an electronic lock **340** that employs a motor **212** configured to rotate an actuator **214**. Rotation of the motor **212** causes the actuator **214** to rotate between a position where the pin **216** is biased to an engaged position with the handle **210** or to a retracted position whereby the handle is released based on loading and unloading of a spring **232**, which in turn causes a shuttle **234** to move linearly. As before, the motor **212** and the pin **216** may be arranged in-line with one another or along the same axis and convert rotational to linear movement.

FIGS. **30** and **31** illustrate a lock mechanism **440** according to another embodiment, which is similar to that described above with respect to FIG. **28** but demonstrates that different types and configurations of lock mechanisms and handles may be employed. In this embodiment, a motor **212** is configured to rotate an actuator **214** for loading or unloading a spring **232** that is engaged with the actuator and a shuttle **234**. Unloading of the spring **232** causes the shuttle **234** to move the pin **216** to an extended position for engaging a latch mechanism **236** to allow rotation of a drive shaft **224** (see, e.g., FIG. **30** where the pin engages a slot defined in the latch mechanism) while loading the spring causes the shuttle to move to a retracted position and out of engagement with the latch mechanism (see, e.g., FIG. **31**), although it is understood that loading or unloading could be used for either extending or retracting the pin based on the direction of rotation of the actuator **214**. In this example, the handle **210'** is configured to rotate when the pin **216** is in an extended and engaged position with the latch mechanism **236** for actuating a latch to an unlocked and disengaged position. When the pin **216** is retracted, rotation of the handle **210'** will not actuate the latch and will not disengage the door. One advantage of this embodiment is that the spring **232** is configured to store energy to be used to ensure that the lock **240** is in the locked or unlocked position as intended. In this way, if the pin **216** is actuated to an extended position but fails to engage the latch mechanism **236** (e.g., due to the handle **210'** being rotated prior to communicating with a key **20**, **120** and actuation of the pin) the spring **232** will store energy and cause the pin to engage the latch mechanism once the handle is rotated back to its initial unlocked position (e.g., so that the pin engages the slot defined in the latch mechanism). In a similar fashion, if one were to attempt to actuate the handle **210'** prior to communicating with a key **20**, **120**, the pin **216** may not retract due to the force being applied between the latch mechanism **236** and the pin; however, once the force is released from the handle, the stored energy in the spring **232** will cause the pin to automatically disengage the latch mechanism. Thus, the lock mechanism **440** in this particular embodiment is configured to store sufficient energy to actuate the lock mechanism without using additional electrical power or a battery.

In some embodiments, the handle **210** is configured to move between an engaged position (e.g., FIG. **23**) and a disengaged position (e.g., FIG. **22**). As shown in FIG. **22**, in the disengaged position, the handle **210** extends outwardly from the housing **241** of the lock **240**. In this way, the operator is able to readily determine that the lock **240** is unlocked, as well as allow the operator to actuate the handle **210** between locked and unlocked positions. For example, the handle **210** may be configured to pivot about one end such that the operator may be able to rotate the handle **210** clockwise or counter-clockwise between locked and unlocked position when the handle has been disengaged with the housing of the lock **240**. In some advantageous embodiments, the handle **210** is configured to automatically disengage and extend from the housing of the lock **240** in response to unlocking of the lock mechanism (e.g., in response to communication with an authorized key **20**, **120** as discussed above). For instance, FIGS. **25**-**27** show an embodiment wherein the lock **240** further includes a rack and pinion mechanism **218** that is configured to cause the handle **210** to pivot about one end to a position extending outwardly from the housing of the lock **240**. In operation, disengagement of the pin **216** causes a rack **220** engaged with the handle **210** to travel along the pinion gear **222**. A spring or the like could be employed to cause the rack **220** to move in response to disengagement of the pin **216**. The pinion gear **222** is fixed in position such that movement of the rack **220** along the pinion gear causes the handle **210** to rotate outwardly (e.g. compare FIGS. **26** and **27**). In one example, the opposite end of the pinion gear **222** may be configured to be attached to the latch **208** such that rotation of the handle **210** rotates the latch. Other mechanisms could be employed to cause the handle **210** to move to a disengaged position, such as one or more springs and/or magnets configured to bias the handle outwardly from the housing of the lock **240**, **340**, **440**.

In some embodiments, techniques for ensuring that the door **206** is closed prior to locking the lock **240**, **340**, **440** and/or ensuring that the lock is locked are provided. In this regard, embodiments may prevent "air locks", which is the instance where the lock **240**, **340**, **440** has been locked, but the door **206** and/or the handle **210** is not actually closed. For example, one or more sensors may be provided for detecting if the door **206** is indeed closed and/or the handle **210** is indeed in the correct position before allowing the lock **240**, **340**, **440** to be activated. Various mechanisms could be used for such detection, such as for example, electronic switches, magnetic detectors, capacitive detectors, light detectors, LED emitters, resistance level detectors, reed switches, optical switches, unique identifiers, and others. In some cases, when the door **206** is closed and the handle **210** is closed, an electrical circuit is completed that then permits the lock **240**, **340**, **440** to be locked. In other words, the lock **240**, **340**, **440** is only able to be locked when the electrical circuit is complete.

In some embodiments, mechanisms may be provided for anti-spoofing protection to protect against unauthorized opening of the lock **240**, **340**, **440**. For instance, the lock **240**, **340**, **440** may employ "smart" detectors such as, for example, detectors configured to detect an expected signal from a key **20**, **120**. In some examples, the detectors could be configured to detect a UPC or QR code or a specific pulsing light or magnetic signals with a code. Such a smart detector could also be configured to determine if tampering of the lock **240**, **340**, **440** had taken place. For example, a plunger switch could detect if the detector had been removed from the lock **240**, **340**, **440** and then provide a notification signal to the lock. In another embodiment, the detector and the lock **240**, **340**, **440** are configured to be paired, so that if an incorrect match is discovered, an alert is generated. Finally, the detector may be configured to read or detect a particular characteristic, such as a magnetic field strength, such that any tampering may change the characteristic and thus indicate a breach had been attempted.

In some embodiments, the lock **240**, **340**, **440** may be configured to provide a final acknowledgement to the key **20**, **120** that it successfully locked. However, if the user pulls the key **20**, **120** away from the lock **240**, **340**, **440** too fast, the acknowledgment may be lost. One example technique to address this problem is provide a lock **240**, **340**, **440** with a power storage device (e.g., a capacitor) that is configured to store sufficient energy to re-open the lock. In other words, when the lock **240**, **340**, **440** locks, the lock provides its acknowledgment and then waits for the key **20**, **120** to respond that the acknowledgment was received. If the lock **240**, **340**, **440** does not receive confirmation from the key **20**, **120**, the lock then unlocks. Thus, the lock **240**, **340**, **440** will only remain locked if a confirmation is received from the key **20**, **120**.

As discussed above, the handle **210** may be configured to automatically lift from the housing of the lock **240**, **340**, **440** when the lock is unlocked. This creates a visual indicator that the handle **210** is not locked. This does not open the door **206**, as the handle **210** has only been moved from its "ready-to-lock" position to its "ready-to-turn" position automatically. In addition, in this embodiment, the location for locking is on the handle **210** (see, e.g., transfer port **242**). In this way, the handle **210** must be in the closed position before the key **20**, **120** is able to communicate with the lock **240**, **340**, **440**. This creates a visual indicator to the operator that the handle **210** must be closed and may also allow one-hand functionality as the key **20**, **120** may itself hold the handle down while locking the lock **240**, **340**, **440**.

In some embodiments, such as for example, those discussed above, a key **20**, **120** may be authorized by a programming station **60**. In some cases, a pin code or other authorization is required to order to authorize a key **20**, **120**. In some server rack facilities, authentication is required just to get into the building storing the racks. Often this is carried out using access cards and/or biometrics. Thus, in some embodiments, the authentication process may be streamlined by using one of the existing methods already implemented in the server rack facility. For example, the existing authentication system may be configured to deliver an authentication signal to the programming station **60** rather than having a user input a separate pin code to indicate that the user is authorized to use the key **20**, **120**. Thus, the programming station **60** may be configured to receive a signal from the local authentication system of the server rack facility. This signal could be delivered using various communication protocols so as to tie the authentication of the user gaining access to the server rack facility to the key **20**, **120** he or she is authenticating. Another embodiment of key authentication is the ability for the system to limit the amount of locks **240**, **340**, **440** a key **20**, **120** is allowed to access. For example, a user might be given a *single* key press to open *one* lock **240**, **340**, **440** and then must return to the programming station **60** to open other locks. Alternately, the reverse could also be programmed such that a given lock **240**, **340**, **440** is only allowed to be opened X times per day and after that, no access is permitted.

Many server racks have different types of mechanical locks from a simple cam, to double throw rods, to sliding multi-latch plates and others. Also, doors **206** have different holes and openings for the lock to attach to. Thus, utilizing a one-size-fits-all lock may be difficult to achieve using the existing footprint of the lock. In some embodiments, mounting features on the lock **240**, **340**, **440** can solve these problems. For example, as shown in FIG. **29**, the drive shaft **224** of the lock **340** that is in engagement with the handle **210** may be flush with the back of the lock and include a keyed socket and/or tapped screw hole **226** or other like attachment point. This configuration allows any variety of adapters and/or latches **208** to be attached to the drive shaft **224** to accommodate different latches and locking mechanisms. Another feature may be a recessed channel **228** that is defined on the back housing of the lock **340** having a variety of attachment mounting points **230** (e.g., fasteners). Thus, various components, such as hooks or plates, can be customized to attach to these attachment points **230** within the recessed channel **228** to adapt the housing of the lock **340** to be attached to any door configuration without causing the dimensions of the lock to change significantly to thereby ensure compatibility with the existing footprint of the lock.

In some embodiments, the lock **240**, **340**, **440** may include a digital display either integrated into, or a module attached to, the lock. This display could have several features such as indicating to the user whether he or she is authorized to open the lock **240**, **340**, **440**. The display may also display a status state (e.g., locked or unlocked), which may be beneficial for ensuring that the racks are secure (e.g., to a security person walking the floor of the server facility to check the status of the locks). The display could indicate various other types of information such as, for example, whether or not the lock **240**, **340**, **440** and door **206** are closed, whether there have been any tamper attempts, and identification of those who accessed that server rack. Maintenance information could also be delivered to the display, such as for technicians working on components in the rack (e.g., for determining which drive is to be replaced).

In some embodiments, various alerts may be provided, such as for detecting concerning situations. Alerts could be audible/visual locally or delivery of a message to an appropriate person or remote device to investigate. Some types of alerts would be tamper attempts or doors not being locked after a certain time limit. More advanced alerts could be implemented as well. For example, if there were standard maintenance times entered into the system (e.g., 20 minutes to remove a drive from a server rack), the system could match the work order to the lock **240**, **340**, **440** opening and then monitor for an aberration of the standard time and then send an alert. Also, technicians could be monitored to see when they are opening racks **240**, **340**, **440**. A long delay between two lock **240**, **340**, **440** openings could indicate an employee taking unauthorized breaks on the job or possibly having time to do something nefarious.

In other embodiments, the key **20**, **120** may be used for ensuring chain of custody. For example, the key **20**, **120** may be configured to scan the rack or hardware contained within the rack (e.g., servers or hard drives). For example, each drive could have an NFC label attached thereto (or any other of a number of devices to be identified), and the key **20**, **120** may be configured to read data on the NFC label. Scanning the NFC label may result in the key **20**, **120** storing information stored on the label which may in turn be stored in the key for auditing purposes. When the technician opens the door **206**, they may also be required to scan the drive they are removing, which could likewise be stored on the key **20**, **120**. In the event the server drives are to be destroyed, the key **20**, **120** may also be configured to scan the drives at the destruction point for storing additional audit data. Thus, the key **20**, **120** can facilitate acquiring more data about when and who accessed a drive, leading to a chain of custody for that drive.

In additional embodiments, the system **200** may include a security device to detect unauthorized access to a server rack **202**. In one example, the security device may be configured to detect removal of a drive contained within the server rack

202. For instance, each drive could have a security device attached to it and then attached to the rack that acts as a "fuse" and if the drive is removed, the fuse is blown. This information can then be delivered to the key **20**, **120** or the lock **240**, **340**, **440** through wired or wireless means. The system **202** may be configured to determine if this was a legitimate removal (e.g., a technician authorized to replace the drive) or an unauthorized removal resulting in sending an alert. Many different techniques could be developed for detecting removal of any component from the server rack, such as for example, a plunger switch, a tether, magnetic sensing, and/or light-based sensing. With respect to fuses, the fuses could also have a detachable mechanism to allow removal without triggering a security event. For example, the same key **20**, **120** that opens the lock **240**, **340**, **440** could be configured to disable the fuse. The data about fuse disablement may also be stored in the key **20**, **120**. Alternately, only certain fuses may be allowed to be disabled by the key **20**, **120** based on the given user and/or the work order. Also, a fuse plugged into a drive (e.g., a cat-5 port) may be configured to deliver an electronic signal to that drive when an unauthorized removal happens—such a signal might be communicated to the drive to erase itself. An unauthorized fuse signal or an unauthorized lock **240**, **340**, **440** opening could also result in sending a signal back to a remote system (e.g., with the key **20**, **120**) to initiate a lock-down whereby no locks **240**, **340**, **440** are allowed to be opened until an override is provided (e.g., by a site manager).

In some embodiments, forced break-ins are sometimes necessary such as when the electronics in the lock **240**, **340**, **440** fails or the lock is mechanically jammed Thus, it may be desirable to include means for differentiating between an authorized break-in due to lock failure and an unauthorized break in. One method of providing such differentiation is to design the lock **240**, **340**, **440** in such a way as to make a break-in attempt obvious. For instance, intentional designs such as thin walls, material selection, or break points could cause the lock **240**, **340**, **440** to fail in such a way that is visually obvious and difficult to cover up. In other examples, notifications could be provided to alert that a forced break-in was attempted. For example, vibration or pressure sensors could be included on the lock **240**, **340**, **440** that are configured to detect anomalous vibrations or pressure and could then send an alert in response to such detection. A number of different sensor types known in the art could accomplish this goal.

The foregoing has described several embodiments of systems, devices, locks, keys, computer storage mediums, and methods. Although embodiments of the present invention have been shown and described, it will be apparent to those skilled in the art that various modifications thereto can be made without departing from the spirit and scope of the invention. Accordingly, the foregoing description is provided for the purpose of illustration only, and not for the purpose of limitation.

That which is claimed is:

1. A security system for a server rack comprising:

an electronic key; and

an electronic lock configured to be attached to the server rack, the electronic lock configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock,

wherein the electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized,

wherein the electronic key comprises an internal power source, and wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source,

wherein the electronic key is only capable of communicating with the electronic lock when the handle is engaged with the electronic lock.

2. The security system of claim **1**, wherein the handle is configured to be actuated for accessing the server rack when the handle is disengaged from the electronic lock, and wherein the electronic lock is configured to re-lock only when the handle is engaged with the electronic lock.

3. The security system of claim **1**, wherein the electronic lock is configured to detect engagement of the handle with the electronic lock.

4. The security system of claim **1**, wherein the handle is configured to be rotated between a latched position and an unlatched position.

5. The security system of claim **1**, wherein the handle is configured to automatically disengage the electronic lock when the electronic key is authorized.

6. The security system of claim **1**, further comprising a programming station for authorizing the programmable electronic key.

7. The security system of claim **4**, wherein the handle is configured to pivot about one end between the latched and unlatched positions.

8. The security system of claim **1**, wherein the electronic lock comprises a mechanism configured to covert rotational movement into linear movement for releasing the handle.

9. The security system of claim **1**, wherein the electronic key is configured to communicate with the electronic lock for re-locking the electronic lock.

10. The security system of claim **9**, wherein the electronic lock is configured to remain re-locked only if a confirmation is received from the electronic key.

11. The security system of claim **1**, wherein the electronic key is configured to obtain identification and/or audit data from the electronic lock.

12. The security system of claim **1**, wherein the electronic key is configured to store identification and/or audit data based on interaction with the electronic lock.

13. The security system of claim **1**, further comprising a plurality of electronic locks, and wherein the electronic key is configured to be authorized for unlocking the plurality of electronic locks.

14. The security system of claim **13**, further comprising a plurality of electronic keys, wherein each of the plurality of electronic keys is configured to be authorized for unlocking different electronic locks.

15. The security system of claim **1**, wherein the electronic key is configured to be authorized if a security code stored by the electronic key matches a security code stored by the electronic lock.

16. A security system for a server rack comprising:

an electronic key;

an electronic lock configured to be attached to the server rack, the electronic lock configured to communicate with the electronic key for determining whether the electronic key is authorized to unlock or lock the electronic lock; and

a latch mechanism operably engaged with the electronic lock and configured to move between a latched position and an unlatched position when the electronic lock is unlocked,

wherein the electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized, wherein the handle is configured to actuate the latch mechanism for accessing the server rack when the handle is disengaged from the electronic lock and the electronic lock is unlocked, and wherein the electronic lock is configured to re-lock only when the handle is engaged with the electronic lock.

17. The security system of claim **16**, wherein the electronic key comprises an internal power source, and wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source to the security device.

18. A method for protecting a server rack from unauthorized access, the method comprising:

communicating between an electronic key and an electronic lock to determine whether the electronic key is authorized to unlock or lock the electronic lock, wherein the electronic lock comprises a handle configured to be disengaged from the electronic lock when the electronic key is authorized, wherein the electronic key is only capable of communicating with the electronic lock when the handle is engaged with the electronic lock; and

transferring electrical power from the internal power source of the electronic key to the security device, wherein the electronic lock is configured to be operated using electrical power transferred from the internal power source if the electronic key is authorized.

19. The method of claim **18**, wherein the electronic lock comprises a handle, and wherein the method further comprises disengaging the handle from the electronic lock when the electronic key is authorized for unlocking the electronic lock and re-locking the electronic lock only when the handle is re-engaged with the electronic lock.

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